Air quality — Certification of automated measuring systems —

Part 3: Performance criteria and test procedures for automated measuring systems for monitoring emissions from stationary sources

ICS 13.040.40

British Standards

National foreword

This British Standard is the UK implementation of EN 15267-3:2007.

The UK participation in its preparation was entrusted by Technical Committee EH/2, Air quality, to Subcommittee EH/2/1, Stationary source emission.

A list of organizations represented on this committee can be obtained on request to its secretary.

This publication does not purport to include all the necessary provisions of a contract. Users are responsible for its correct application.

Compliance with a British Standard cannot confer immunity from legal obligations.

This British Standard was published under the authority of the Standards Policy and Strategy Committee on 31 March 2008

ISBN 978 0 580 55617 3

Amendments/corrigenda issued since publication

Date	Comments

EUROPEAN STANDARD
NORME EUROPÉENNE
EUROPÄISCHE NORM

EN 15267-3

December 2007

ICS 13.040.40

English Version

Air quality - Certification of automated measuring systems - Part 3: Performance criteria and test procedures for automated measuring systems for monitoring emissions from stationary sources

Qualité de l'air - Certification des systèmes de mesurage automatisés - Partie 3: Spécifications de performance et procédures d'essai pour systèmes de mesurage automatisés des émissions de sources fixes

Luftbeschaffenheit - Zertifizierung von automatischen Messeinrichtungen - Teil 3: Mindestanforderungen und Prüfprozeduren für automatische Messeinrichtungen zur Überwachung von Emissionen aus stationären Quellen

This European Standard was approved by CEN on 17 November 2007.

CEN members are bound to comply with the CEN/CENELEC Internal Regulations which stipulate the conditions for giving this European Standard the status of a national standard without any alteration. Up-to-date lists and bibliographical references concerning such national standards may be obtained on application to the CEN Management Centre or to any CEN member.

This European Standard exists in three official versions (English, French, German). A version in any other language made by translation under the responsibility of a CEN member into its own language and notified to the CEN Management Centre has the same status as the official versions.

CEN members are the national standards bodies of Austria, Belgium, Bulgaria, Cyprus, Czech Republic, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Iceland, Ireland, Italy, Latvia, Lithuania, Luxembourg, Malta, Netherlands, Norway, Poland, Portugal, Romania, Slovakia, Slovenia, Spain, Sweden, Switzerland and United Kingdom.

EUROPEAN COMMITTEE FOR STANDARDIZATION
COMITÉ EUROPÉEN DE NORMALISATION
EUROPÄISCHES KOMITEE FÜR NORMUNG

Management Centre: rue de Stassart, 36 B-1050 Brussels

Ref. No. EN 15267-3:2007: E

Contents

Page

Foreword

This document (EN 15267-3:2007) has been prepared by Technical Committee CEN/TC 264 “Air quality”, the secretariat of which is held by DIN.

This document shall be given the status of a national standard, either by publication of an identical text or by endorsement, at the latest by June 2008 and conflicting national standards shall be withdrawn at the latest by June 2008.

This document is Part 3 of a series of European Standards:

— EN 15267-1, *Air quality — Certification of automated measuring systems — Part 1: General principles*

— EN 15267-2, *Air quality — Certification of automated measuring systems — Part 2: Initial assessment of the AMS manufacturer’s quality management system and post certification surveillance for the manufacturing process*

— EN 15267-3, *Air quality — Certification of automated measuring systems — Part 3: Performance criteria and test procedures for automated measuring systems for monitoring emissions from stationary sources*

— EN 15267-4, *Air quality — Certification of automated measuring systems — Part 4: Performance criteria and test procedures for automated measuring systems for monitoring ambient air quality*

According to the CEN/CENELEC Internal Regulations, the national standards organizations of the following countries are bound to implement this European Standard: Austria, Belgium, Bulgaria, Cyprus, Czech Republic, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Iceland, Ireland, Italy, Latvia, Lithuania, Luxembourg, Malta, Netherlands, Norway, Poland, Portugal, Romania, Slovakia, Slovenia, Spain, Sweden, Switzerland and United Kingdom.

0 Introduction

0.1 General

CEN has established standards for the certification of automated measuring systems (AMS) used for monitoring emissions from stationary sources and ambient air quality. This product certification is based on the following four sequential stages:

a) performance testing of an AMS;

b) initial assessment of the AMS manufacturer's quality management system;

c) certification;

d) post certification surveillance.

This European Standard defines the performance criteria and test procedures for performance testing of AMS used to monitor emissions from stationary sources. Testing applies to complete measuring systems.

The overall assessment for the purposes of certification is *conformity testing*, whilst the evaluation of performance against specified performance criteria is *performance testing*.

0.2 Legal drivers

This European Standard supports the requirements of the following EU Directives:

— Directive on the limitation of emissions of certain pollutants into the air from large combustion plants (2001/80/EC);

— Directive on the incineration of waste (2000/76/EC);

— Directive on the limitation of emissions of volatile organic compounds due to the use of organic solvents in certain activities and installations (1999/13/EC);

— Integrated Pollution Prevention and Control Directive (1996/61/EC);

— Directive on processes emitting greenhouse gases (2003/87/EC).

However, this European Standard can also be applied to the monitoring requirements specified in other EU Directives.

0.3 Relationship to EN 14181

The Quality Assurance Levels (QAL) defined in EN 14181 cover the suitability of an AMS for its measuring task (QAL1), the regular calibration and validation of the AMS (QAL2), and the control of the AMS during its ongoing operation on an industrial plant (QAL3). An Annual Surveillance Test (AST) is also defined in EN 14181.

This European Standard provides the detailed procedures covering the QAL1 requirements of EN 14181. Furthermore, it provides input data for QAL3.

0.4 Processes

Field testing of an AMS is ordinarily carried out on the most highly demanding industrial process in the range of applications for which a manufacturer seeks certification. The premise is that if the AMS performs acceptably on this process, then experience has shown that the AMS generally performs well on the majority of other processes. However, there are always exceptions and it is the responsibility of the manufacturer in conjunction with the user to ensure that the AMS performs adequately on a specific process.

0.5 Performance characteristics

A combination of laboratory and field testing is detailed within this European Standard. Laboratory testing is designed to assess whether an AMS can meet, under controlled conditions, the relevant performance criteria. Field testing, over a minimum three month period, is designed to assess whether an AMS can continue to work and meet the relevant performance criteria in a real application. Field testing is carried out on an industrial process representative of the intended application for the AMS for which the manufacturer seeks certification.

The main AMS performance characteristics are

— response time,

— repeatability standard deviation at zero and span points,

— lack of fit (linearity) under laboratory and field conditions,

— zero and span drift under laboratory and field conditions,

— influence of ambient temperature,

— influence of sample gas pressure,

— influence of sample gas flow for extractive AMS,

— influence of voltage variations,

— influence of vibration,

— cross-sensitivity to likely interferents contained in the waste gas other than the measured component,

— excursion of measurement beam of in-situ AMS,

— converter efficiency for NO_x AMS,

— response factors,

— performance and accuracy of the AMS against a standard reference method (SRM) under field conditions,

— maintenance interval under field conditions,

— availability under field conditions and

— reproducibility under field conditions.

The quality of reference or surrogate materials used under QAL3 for particulate matter measuring AMS is also assessed.

This European Standard is an application and elaboration of EN ISO 9169 with additional and alternative provisions for paired testing. Where this European Standard appears to differ from EN ISO 9169, it either elaborates upon the requirements of EN ISO 9169 or differs in minor ways owing to the necessity to conduct paired testing.

1 Scope

This European Standard specifies the performance criteria and test procedures for automated measuring systems that measure gases and particulate matter in, and flow of, the waste gas from stationary sources.

This European Standard supports the requirements of particular EU Directives. It provides the detailed procedures covering the QAL1 requirements of EN 14181 and, where required, input data used in QAL3.

2 Normative references

The following referenced documents are indispensable for the application of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

EN 12619, *Stationary source emissions — Determination of the mass concentration of total gaseous organic carbon at low concentrations in flue gases — Continuous flame ionisation detector method*

EN 13284-1, *Stationary source emissions — Determination of low range mass concentration of dust — Part 1: Manual gravimetric method*

EN 13284-2, *Stationary source emissions — Determination of low range mass concentration of dust — Part 2: Automated measuring systems*

EN 13526, *Stationary source emissions — Determination of the mass concentration of total gaseous organic carbon in flue gases from solvent using processes — Continuous flame ionisation detector method*

EN 14181:2004, *Stationary source emissions — Quality assurance of automated measuring systems*

EN 15259:2007, *Air quality — Measurement of stationary source emissions — Requirements for measurement sections and sites and for the measurement objective, plan and report*

EN 50160, *Voltage characteristics of electricity supplied by public distribution systems*

EN 60529, *Degrees of protection provided by enclosures (IP code) (IEC 60529:1989)*

EN 60068-1, *Environmental testing — Part 1: General and guidance (IEC 60068-1:1988 + Corrigendum 1988 + A1:1992)*

EN 60068-2 (all tests), *Environmental testing — Part 2: Tests*

EN ISO 14956, *Air quality — Evaluation of the suitability of a measurement procedure by comparison with a required measurement uncertainty (ISO 14956:2002)*

EN ISO/IEC 17025, *General requirements for the competence of testing and calibration laboratories (ISO/IEC 17025:2005)*

3 Terms and definitions

For the purposes of this document, the following terms and definitions apply.

3.1
automated measuring system
AMS
entirety of all measuring instruments and additional devices for obtaining a result of measurement

NOTE 1 Apart from the actual measuring device (the analyser), an AMS includes facilities for taking samples (e.g. probe, sample gas lines, flow meters and regulator, delivery pump) and for sample conditioning (e.g. dust filter, pre-separator for interferents, cooler, converter). This definition also includes testing and adjusting devices that are required for functional checks and, if applicable, for commissioning.

NOTE 2 The term “automated measuring system” (AMS) is typically used in Europe. The term “continuous emission monitoring system” (CEM) is also typically used in the UK and USA.

3.2
reference method
RM
measurement method taken as a reference by convention, which gives, the accepted reference value of the measurand

NOTE 1 A reference method is fully described.

NOTE 2 A reference method can be a manual or an automated method.

NOTE 3 Alternative methods can be used if equivalence to the reference method has been demonstrated.

[EN 15259:2007, 3.8]

3.3
standard reference method
SRM
reference method prescribed by European or national legislation

NOTE Standard reference methods are used e.g. to calibrate and validate AMS and for periodic measurements to check compliance with limit values.

[EN 15259:2007, 3.9]

3.4
measurement
set of operations having the object of determining a value of a quantity

[VIM:1993, 2.1]

3.5
paired measurement
simultaneous recording of results of measurement at the same measurement point using two AMS of identical design

3.6
measurand
particular quantity subject to measurement

[VIM:1993, 2.6]

NOTE The measurand is a quantifiable property of the waste gas under test, for example mass concentration of a measured component, temperature, velocity, mass flow, oxygen content and water vapour content.

3.7
measured component
constituent of the waste gas for which a defined measurand is to be determined by measurement

[EN 15259:2007, 3.6]

NOTE Measured component is also called determinand.

3.8
interferent
substance present in the waste gas under investigation, other than the measured component, that affects the output

3.9
calibration
determination of a calibration function with (time) limited validity applicable to an AMS at a specific measurement site

3.10
calibration function
linear relationship between the values of the SRM and the AMS with the assumption of a constant residual standard deviation

[EN 14181:2004, 3.3]

NOTE The calibration function describes the statistical relationship between the starting variable (measured signal) of the measuring system and the associated result of measurement (measured value) simultaneously determined at the same point of measurement using a SRM.

3.11
reference material
substance or mixture of substances, with a known concentration within specified limits, or a device of known characteristics

3.12
zero gas
gas mixture used to establish the zero point of a calibration curve when used with a given analytical procedure within a given calibration range

3.13
zero point
specified value of the output quantity (measured signal) of the AMS and which, in the absence of the measured component, represents the zero crossing of the AMS characteristic

NOTE In case of oxygen and some flow monitoring AMS, the zero point is interpreted as the lowest measurable value.

3.14
span point
value of the output quantity (measured signal) of the AMS for the purpose of calibrating, adjusting, etc. that represents a correct measured value generated by reference material between 70 % and 90 % of the range tested

3.15
measured signal
output from an AMS in analogue or digital form which is converted into the measured value with the aid of the calibration function

3.16
output
reading, or digital or analogue electrical signal generated by an AMS in response to a measured object

3.17
independent reading
reading that is not influenced by a previous individual reading by separating two individual readings by at least four response times

3.18
individual reading
reading averaged over a time period equal to the response time of the AMS

3.19
performance characteristic
quantity assigned to an AMS in order to define its performance

NOTE The values of relevant performance characteristics are determined in the performance testing and compared to the applicable performance criteria.

3.20
accuracy
closeness in agreement between a single measured value of the measurand, and the true value (or an accepted reference value)

3.21
availability
fraction of the total monitoring time for which data of acceptable quality have been collected

3.22
averaging time
period of time over which an arithmetic or time-weighted average of concentrations is calculated

3.23
converter efficiency
efficiency with which the converter unit of a NO_x analyser reduces NO_2 to NO

3.24
interference
negative or positive effect that a substance has upon the output of the AMS, when that substance is not the measured component

3.25
cross-sensitivity
response of the AMS to interferents

NOTE See interference.

3.26
drift
monotonic change of the calibration function over a stated period of unattended operation, which results in a change of the measured value

3.27
zero drift
change in the AMS reading at the zero point over the maintenance interval

3.28
span drift
change in AMS reading at the span point over the maintenance interval

3.29
maintenance interval
maximum admissible interval of time for which the performance characteristics remain within a pre-defined range without external servicing, e.g. refill, calibration, adjustment

NOTE This is also known as the period of unattended operation.

3.30
lack of fit
systematic deviation, within the range of application, between the accepted value of a reference material applied to the measuring system and the corresponding result of measurement produced by the calibrated measuring system

NOTE In common language lack of fit is often called "linearity" or "deviation from linearity". Lack of fit test is often called "linearity test".

3.31
response time
t_{90}
time interval between the instant of a sudden change in the value of the input quantity to an AMS and the time as from which the value of the output quantity is reliably maintained above 90 % of the correct value of the input quantity

NOTE The response time is also referred to as the 90 % time.

3.32
repeatability
ability of an AMS to provide closely similar indications for repeated applications of the same measurand under the same conditions of measurement

3.33
reproducibility
R_f
measure of the agreement between two identical measuring systems applied in parallel in field tests at a level of confidence of 95 % using the standard deviation of the difference of the paired measurements

NOTE 1 Reproducibility is determined by means of two identical AMS operated side by side. It is an AMS performance characteristic for describing the production tolerance specific to that AMS. The reproducibility is calculated from the half-hour averaged output signals (raw values as analogue or digital outputs) during the three-month field test.

NOTE 2 The term "field repeatability" is sometimes used instead of reproducibility.

3.34
uncertainty
parameter associated with the result of a measurement, that characterises the dispersion of the values that could reasonably be attributed to the measurand

[ENV 13005:1999, B.2.18]

3.35
standard uncertainty
uncertainty of the result of measurement expressed as a standard deviation

[ENV 13005:1999, 2.3.1]

3.36
expanded uncertainty
quantity defining an interval about the result of a measurement that may be expected to encompass a large fraction of the distribution of values that could reasonably be attributed to the measurand

[ENV 13005:1999, 2.3.5]

NOTE The interval about the result of measurement is established for a level of confidence of typically 95 %.

3.37
test laboratory
laboratory accredited to EN ISO/IEC 17025 for carrying out performance tests

NOTE CEN/TS 15675 provides an elaboration of EN ISO/IEC 17025 for application to emission measurements, which should be followed when using standard reference methods listed in Annex A.

3.38
field test
test for at least three months on a plant appropriate to the field of application of the AMS

3.39
certification range
range over which the AMS is tested and certified for compliance with the relevant performance criteria

NOTE Certification range is always related to the daily ELV.

3.40
emissions limit value
ELV
limit value given in regulations such as EU Directives, ordinances, administrative regulations, permits, licences, authorisations or consents

NOTE ELV can be stated as concentration limits expressed as half-hourly, hourly and daily averaged values, or mass flow limits expressed as hourly, daily, weekly, monthly or annually aggregated values.

3.41
plant
installation
industrial facility on which an AMS is installed

4 Symbols and abbreviations

For the purposes of this document, the following symbols and abbreviations apply.

4.1 Symbols

a average value of the AMS readings in the linearity test

A intercept of the regression function in the linearity test

b_f sensitivity coefficient of sample gas flow

b_p sensitivity coefficient of sample gas pressure

b_t sensitivity coefficient of ambient temperature

b_v sensitivity coefficient of supply voltage

B slope of the regression function in the linearity test

c concentration; value of the reference material

c_i carbon mass concentration of substance i at 273 K and 1 013 hPa; individual reference material value

c_ref carbon mass concentration of propane at 273 K and 1 013 hPa

$c_{\mathrm{NO},0}$	concentration of NO with ozone generator switched-off
$c_{\mathrm{NO},i}$	concentration of NO with ozone generator at setting i (i = 1 to n)
$c_{\mathrm{NOx},0}$	concentration of total NO_x with ozone generator switched-off
$c_{\mathrm{NOx},i}$	concentration of total NO_x with ozone generator at setting i (i = 1 to n)
$\overline{c}$	average of c values
d_c	residual
$d_{c,\mathrm{rel}}$	relative residual
E_i	converter efficiency at setting i of the ozone generator (i = 1 to n)
f_i	carbon-related response factor for substance i
m_c	number of repetitions at reference material level c
n	number of measurements; number of parallel measurements
p_1	lower sample gas pressure
p_2	higher sample gas pressure
Δp	difference in sample gas pressure
r_1	nominal flow rate
r_2	lowest flow rate specified by the manufacturer
R_f	reproducibility under field conditions
R	regression coefficient of calibration function
R^2	determination coefficient of calibration function
s_D	standard deviation from paired measurements
s_r	repeatability standard deviation of the measurement
$t_{n-1;\,0,95}$	two-sided Students t-factor at a confidence level of 95 % with a number of degrees of freedom $n - 1$
t_d	relative difference between the response times determined in rise and fall mode
t_r	response time determined in rise mode (average of four measurements)
t_f	response time determined in fall mode (average of four measurements)
t_o	outage time
t_tot	total operating time
T	temperature (absolute)
T_i	ith temperature

u_c combined standard uncertainty

u_{ce} uncertainty contribution caused by converter efficiency for AMS measuring NO_x

$u_{d,s}$ uncertainty contribution caused by span drift from field test

$u_{d,z}$ uncertainty contribution caused by zero drift from field test

u_D uncertainty contribution caused by standard deviation from paired measurements under field conditions

u_f uncertainty contribution caused by influence of sample gas flow

u_i uncertainty contribution caused by cross-sensitivity (interference)

u_i uncertainty contribution to the total uncertainty of the measured values caused by a variation of influence quantity X_i

u_{lof} uncertainty contribution caused by lack of fit

u_{mb} uncertainty contribution caused by excursion of measurement beam

u_p uncertainty contribution caused by influence of sample gas pressure

u_r uncertainty contribution caused by repeatability standard deviation at span

u_{rf} uncertainty contribution caused by variation of response factors (TOC)

u_{rm} uncertainty contribution caused by reference material provided by the manufacturer

u_t uncertainty contribution caused by influence of ambient temperature at span

u_v uncertainty contribution caused by influence of supply voltage

$u(X_i)$ standard uncertainty of the influence quantity X_i

U_1 minimum voltage specified by the manufacturer

U_2 maximum voltage specified by the manufacturer

$U_{0,95}$ expanded uncertainty at a level of confidence of 95 %

V availability

x measured signal

x_i ith measured signal; average of the measured signals for substance i

$x_{i,\text{min}}$ minimum value of the average reading influenced by performance characteristic i during the performance test

$x_{i,\text{max}}$ maximum value of the average reading influenced by performance characteristic i during the performance test

$x_{i,\text{adj}}$ value of the average reading with the influence quantity at its nominal value during the performance test

$x_{1,i}$ ith measured signal of the first measuring system

$x_{2,i}$ — ith measured signal of the second measuring system

$x_{c,i}$ — individual AMS reading at reference material level c

x_{ref} — average of the measured signals for propane

x_{u} — upper limit of the certification range

$\bar{x}$ — average of measured signals x_i

$\bar{x}_c$ — average AMS reading at reference material level c

X_i — ith influence quantity

4.2 Abbreviations

AMS — Automated Measuring System

AST — Annual Surveillance Test

ELV — Emission Limit Value

QAL — Quality Assurance Level

QAL1 — First Quality Assurance Level

QAL2 — Second Quality Assurance Level

QAL3 — Third Quality Assurance Level

SRM — Standard Reference Method

TOC — Total Organic Carbon

5 General requirements

5.1 Application of performance criteria

The test laboratory shall test at least two identical automated measuring systems (AMS). All AMS tested shall meet the performance criteria specified in this document as well as the uncertainty requirement specified in the applicable regulations.

5.2 Ranges to be tested

5.2.1 Certification range

The certification range over which the AMS is to be tested shall comprise minimum and maximum values. The coverage shall be fit for the intended application of the AMS. The certification range shall be specified as follows:

a) for waste incinerators as the range usually begins from zero, if the AMS is able to measure zero, and a value no greater than 1,5 times the daily average emissions limit value (ELV);

b) for large combustion plants as the range usually begins from zero, if the AMS is able to measure zero, and a value no greater than 2,5 times the daily average emissions limit value (ELV);

c) for other plants in relation to the corresponding emission limit value or any other requirement related to the intended application.

The AMS shall be able to measure instantaneous values in a range that is at least 2 times the upper limit of the certification range in order to measure the half-hour values. If it is necessary to use more than one range setting of the AMS to achieve this requirement, these supplementary ranges will require additional testing (see 5.2.2).

NOTE 1 In addition to the certification ranges stated above, manufacturers can choose supplementary ranges that are larger than the certification range.

NOTE 2 Manufacturers can choose other ranges for different applications. If an AMS is tested e.g. for use on waste incinerators, it can also be used on large combustion plants, if the supplementary ranges are tested as given in 5.2.2.

The certification range(s) and the performance criteria tested for each range shall be stated on the certificate.

The test laboratory should choose for the field test an industrial plant with challenging measuring conditions. This means that the AMS can also be used under less demanding measuring conditions.

5.2.2 Supplementary ranges

If a manufacturer wishes to demonstrate performance over one or more supplementary ranges larger than the certification range, some limited additional testing is required over all the supplementary ranges. This additional testing shall at least include evaluations of the response time (see 10.9) and lack of fit (see 10.12). Cross-sensitivity (see 10.19) has to be tested for interferents that have shown relevance during testing in the certification range. The concentration of the relevant interferents shall be proportionally higher than the values specified in Table B.1, where the proportionality factor is given by the ratio of the considered supplementary range to the certification range.

Supplementary ranges and the performance criteria tested for these ranges shall be stated on the certificate.

5.2.3 Lower limit of ranges

The lower limit of the certification range is usually zero.

NOTE 1 The zero value is typically the detection limit.

NOTE 2 For oxygen measuring AMS the lower limit of the certification range can differ from zero.

5.2.4 Expression of performance criteria with respect to ranges

The performance criteria presented in Clause 6 are expressed in terms of a percentage of the upper limit of the certification range for each measured component except for oxygen where the performance criteria are expressed as volume concentrations. A performance criterion with respect to ranges is a value that corresponds to the largest deviation allowed for each test, regardless of the sign of the deviation determined in the test.

5.2.5 Ranges of optical in-situ AMS with variable optical length

The certification range for optical in-situ AMS with variable optical length shall be defined in units of the measured component concentration multiplied by the length of the optical path.

The path length used for testing shall be stated on the certificate.

5.3 Manufacturing consistency and changes to AMS design

Certification is specific to the AMS version that has undergone performance testing. Subsequent design modifications that might affect the performance of the AMS can invalidate the certification.

NOTE Design modifications apply to both hardware and software.

Manufacturing consistency and changes to AMS design are described in EN 15267-2.

5.4 Qualifications of test laboratories

Test laboratories shall be accredited to EN ISO/IEC 17025 and the appropriate test standards for carrying out the tests defined in this European Standard. Test laboratories shall have knowledge on the uncertainties attributed to the individual test procedures applied during performance testing.

CEN/TS 15675 provides an elaboration of EN ISO/IEC 17025 for application to emission measurements which should be followed when using standard reference methods specified in Annex A.

6 Performance criteria common to all AMS for laboratory testing

6.1 AMS for testing

All AMS submitted for testing shall be complete. These specifications do not apply to the individual parts of an AMS. The test report shall be issued for a specified AMS with all its parts listed.

An AMS that uses extractive sampling systems shall have appropriate provisions for filtering solids, avoiding chemical reactions within the sampling system, entrainment effects and effective control of water condensate.

Measuring systems with different options for the sampling line length shall be tested with an appropriate sampling line length agreed between the test laboratory and the manufacturer. The length shall be quoted in the test report.

NOTE Use of longer sampling lines is covered by QAL2.

The test laboratory shall describe in the test report the type of sampling system.

6.2 CE labelling

The AMS shall comply with the requirements for CE labelling specified in applicable EU Directives. These include, for example:

— Electro-Magnetic Compatibility Directive 89/336/EEC and its amendments 92/31/EEC and 93/68/EEC, and

— Low-Voltage Directive 72/23/EEC and its amendment 93/68/EEC covering electrical equipment designed for use within certain voltage limits.

AMS manufacturers or suppliers shall supply verifiable and traceable evidence of compliance with the requirements of the relevant EU Directives applicable to the equipment.

6.3 Security

The AMS shall have a means of protection against unauthorised access to control functions.

6.4 Output ranges and zero-point

The AMS shall have a data output with a living zero point (e.g. 4 mA) so that both negative and positive readings can be displayed.

The AMS shall have a display that shows the measurement response. The display may be external to the AMS.

6.5 Additional data outputs

The AMS shall have a data output allowing an additional data display and recording device to be fitted to the AMS, i.e. one for the data acquisition system and one supplementary output for QAL2, QAL3 and AST according to EN 14181.

6.6 Display of operational status signals

The AMS shall have a means of displaying its operating status.

NOTE Status signals cover, for example, normal operation, stand-by, maintenance mode and malfunctions.

The AMS shall also have a means of communicating the operational status to a data handling and acquisition system.

6.7 Prevention or compensation for optical contamination

An AMS that uses an optical method as the measuring principle shall have provisions for either prevention of contamination of the optical system and/or compensation for its effects.

6.8 Degrees of protection provided by enclosures

Instruments limited to be mounted in ventilated rooms or cabinets, where any kind of precipitation cannot reach the instrument, shall meet at least IP40 as specified in EN 60529.

Instruments limited to being mounted in areas where some kind of shelter against precipitation is in place, e.g. a porch roof, but where precipitation can reach the instrument due to wind etc., shall meet at least IP54 as specified in EN 60529.

Instruments that are designed to be used in the open air and without any weather protection shall at least meet the requirements of standard IP65 as specified in EN 60529.

6.9 Response time

The AMS shall meet the performance criteria for response time specified in Clause 8.

6.10 Repeatability standard deviation at zero point

The AMS shall meet the performance criteria for repeatability standard deviation at the zero point specified in Clause 8.

NOTE 1 The detection limit is two times the repeatability standard deviation at zero.

NOTE 2 The quantification limit is four times the repeatability standard deviation at zero.

6.11 Repeatability standard deviation at the span point

The AMS shall meet the performance criteria for repeatability standard deviation at the span point specified in Clause 8.

6.12 Lack of fit

The AMS shall have a linear output and shall meet the performance criteria for lack of fit specified in Clause 8.

6.13 Zero and span drift

The manufacturer shall provide a description of the technique used by the AMS to determine and compensate the zero and span drift. The description shall not be limited to an explanation of how the AMS compensates for the effect of contamination of the optical surfaces of AMS that use optical techniques.

The test laboratory shall assess that the chosen reference material applied to the AMS as an independent check of the instrument's operation is capable of monitoring any relevant change in instrument response not caused by changes in the measured component or stack gas condition.

The AMS shall allow recording the zero and span drift. The manufacturer shall describe how to obtain the zero and span values.

The technique used should be sensitive to drift in as many of the active parts of the system as possible.

If the AMS has a means of automatic compensation for contamination and calibration and re-adjustment for zero and span drift, and such adjustments are not capable of bringing the AMS within normal operational conditions, then the AMS shall set a status signal.

In cases where the AMS cannot measure zero values, the drift has to be measured at the lower limit of the certification range.

NOTE For example, some AMS which measure flow and oxygen are not able to measure zero values.

6.14 Influence of ambient temperature

The deviations of the AMS readings at the zero and span points shall not exceed the performance criteria specified in Clause 8 for the following test ranges of the ambient temperature:

— from –20 °C to +50 °C for assemblies installed outdoors;

— from +5 °C to +40 °C for assemblies installed indoors, where the temperatures do not fall below +5 °C or rise above +40 °C.

The manufacturer submitting an AMS for testing may specify wider ambient temperature ranges to those above.

NOTE Temperature ranges tested are indicated in the certificate.

6.15 Influence of sample gas pressure

The deviations of the AMS reading at the span point shall not exceed the performance criterion specified in Clause 8 when the sample gas pressure changes by 3 kPa above and below atmospheric pressure.

NOTE This typically applies to in-situ AMS, but not to extractive AMS, since the sample gas is conditioned and typically not subject to significant variations of temperature and pressure once within the analyser.

6.16 Influence of sample gas flow for extractive AMS

The deviations of the AMS reading at the zero point and span point shall not exceed the performance criterion specified in Clause 8, when the sample gas flow is changed in accordance with the manufacturer's specification.

A status signal for the lower limit of the sample gas flow shall be provided.

6.17 Influence of voltage variations

The deviations of the AMS reading at the zero and span points shall not exceed the performance criterion specified in Clause 8 when the voltage supply to the AMS varies from –15 % from the nominal value below to +10 % from the nominal value above the nominal value of the supply voltage.

The AMS shall be capable of operating at a voltage that meets the requirements of EN 50160.

6.18 Influence of vibration

The deviations of the AMS readings at the zero and span points caused by vibrations typically expected at an industrial plant shall not exceed the performance criteria specified in Clause 8.

6.19 Cross-sensitivity

The manufacturer shall describe any known sources of interference. Tests for non-gaseous interference sources, or gases other than those listed in Annex B, shall be agreed with the test laboratory.

The AMS shall meet the performance criteria at the zero and span point for cross-sensitivity specified in Clause 8.

6.20 Excursion of measurement beam of in-situ AMS

In the event of an excursion of the measurement beam within an AMS, the deviations of the AMS readings at the zero point and span point shall not exceed the performance criterion specified in Clause 8 for the maximum allowable deviation angle specified by the manufacturer. This angle shall not be smaller than 0,3 °.

6.21 Converter efficiency for NO_x AMS

Manufacturers shall specify, when seeking certification for AMS for measuring NO_x, whether certification is required for the measurement of nitrogen monoxide (NO) and/or nitrogen dioxide (NO_2). If a converter is used, the converter shall meet the performance criteria for the converter efficiency specified in Clause 8.

NOTE 1 NO_x ordinarily means nitrogen monoxide (NO) plus nitrogen dioxide (NO_2).

NOTE 2 NO_x concentrations are generally expressed as NO_2.

6.22 Response factors

AMS for TOC shall meet the performance criteria specified in Clause 8.

7 Performance criteria common to all AMS for field testing

7.1 Calibration function

The calibration function shall be determined by parallel measurements carried out using a SRM.

NOTE 1 If the concentration in the field does not vary, the calibration function can be established in accordance with EN 14181 by additional use of zero and span values obtained in the field test.

NOTE 2 The case of quadratic calibration functions is described in EN 13284-2.

The calibration function shall have a determination coefficient R^2 of the regression of at least 0,90.

NOTE 3 The determination coefficient R^2 is the square of the regression coefficient R.

The variability attached to the calibration function and determined in accordance with EN 14181 shall meet the maximum permissible uncertainty specified by the applicable regulations.

7.2 Response time

The AMS shall meet the performance criterion for the response time evaluated during the laboratory tests.

NOTE The test for the response time is repeated during the field test, as field conditions can influence the response time.

7.3 Lack of fit

The AMS shall meet the performance criterion for lack of fit evaluated during the laboratory tests.

NOTE The test for the lack of fit is repeated during the field test, as field conditions can influence the lack of fit.

7.4 Maintenance interval

The minimum maintenance interval of the AMS shall meet the performance criterion specified in Clause 8.

7.5 Zero and span drift

The zero and span drift within the maintenance interval shall not exceed the performance criteria specified in Clause 8.

The span materials (such as test gases) applied during testing shall produce an AMS response between 70 % and 90 % of the upper limit of the certification range.

NOTE As field conditions can influence drift behaviour, tests for this characteristic are repeated during the field test.

7.6 Availability

The AMS shall have an availability that meets the requirements of applicable regulations and in any case, the performance criterion specified in Clause 8 during the field test.

NOTE If the AMS is not measuring the measured component for any reason, then it is not considered available for measurements. The AMS can be unavailable due to malfunctions, servicing and any kind of zero and span point evaluation and correction. Periods when the monitored process is not operating are excluded.

7.7 Reproducibility

AMS shall meet the performance criterion for reproducibility under field conditions specified in Clause 8.

7.8 Contamination check of in-situ systems

The response of the AMS to soiling shall be determined in the field test by means of visual checks and, for example, by determining the deviations from the nominal values of the AMS output signal.

If required, the AMS shall be provided with recommended air purging systems for three months as part of the field test. At the end of the test, the effect of the contamination shall be evaluated. The results with clean and soiled optical surfaces shall differ by no more than 2 % of the upper limit of the certification range.

8 Performance criteria specific to measured components

8.1 General

Clause 8 defines the performance criteria for AMS specific to measured components. The values for individual parameters given in these sections are expressed as a percentage of the upper limit of the certification range of the AMS under test, with the exception of availability and calibration function.

Where regulations specify uncertainty requirements, the AMS shall meet both the individual performance criteria specified in this document and the uncertainty requirements required by the applicable regulations. The uncertainty budget shall be determined using the procedure described in Annex E.

8.2 Gas monitoring AMS

8.2.1 Performance criteria

AMS for measuring gaseous measured components shall meet the performance criteria specified in Table 1 and Table 2. The maximum allowable deviations (as absolute values) of the measured signals are given as volume concentration (volume fraction) for oxygen measuring AMS and as percentages of the upper limit of the certification range for other gases.

For AMS which measure moisture as a means of providing data corrected to dry conditions, moisture shall be included as a measured component and the AMS shall meet the performance criteria in Table 1 and Table 2.

Table 1 shows the performance criteria, which are tested in the laboratory. Table 2 shows the performance criteria, which are tested during the three month field test.

Table 1 — Performance criteria for gas monitoring AMS in laboratory tests

Performance characteristic	Performance criteria		Test in sub-clause
	Gases except O_2	O_2	
Response time	≤ 200 s ≤ 400 s for NH_3, HCl and HF	≤ 200 s	10.9
Repeatability standard deviation at zero point	≤ 2,0 % [a]	≤ 0,20 % [b]	10.10
Repeatability standard deviation at span point	≤ 2,0 % [a]	≤ 0,20 % [b]	10.11
Lack of fit	≤ 2,0 % [a]	≤ 0,20 % [b]	10.12
Influence of ambient temperature change from nominal value at 20 °C within specified range at zero point	≤ 5,0 % [a]	≤ 0,50 % [b]	10.14
Influence of ambient temperature change from nominal value at 20 °C within specified range at span point	≤ 5,0 % [a]	≤ 0,50 % [b]	10.14
Influence of sample gas pressure at span point, for a pressure change Δp of 3 kPa	≤ 2,0 % [a]	≤ 0,20 % [b]	10.15
Influence of sample gas flow on extractive AMS for a given specification by the manufacturer	≤ 2,0 % [a]	≤ 0,20 % [b]	10.16
Influence of voltage, at –15 % below and at +10 % above nominal supply voltage	≤ 2,0 % [a]	≤ 0,20 % [b]	10.17
Influence of vibration	≤ 2,0 % [a]	≤ 0,20 % [b]	10.18
Cross-sensitivity	≤ 4,0 % [a]	≤ 0,40 % [b]	10.19
Excursion of the measurement beam of in-situ AMS	≤ 2,0 % [a]	–	10.20
Converter efficiency for AMS measuring NO_x	≥ 95,0 %	–	10.21

[a] Percentage value as percentage of the upper limit of the certification range.

[b] Percentage value as oxygen volume concentration (volume fraction).

Table 2 — Performance criteria for gas monitoring AMS in field tests

Performance characteristic	Performance criteria		Test in sub-clause
	Gases except O_2	O_2	
Determination coefficient of calibration function, R^2	≥ 0,90	≥ 0,90	12.1
Response time	≤ 200 s ≤ 400 s for NH_3, HCl and HF	≤ 200 s	12.2
Lack of fit	≤ 2,0 % [a]	≤ 0,20 % [b]	12.4
Minimum maintenance interval	8 days	8 days	12.4
Zero drift within maintenance interval	≤ 3,0 % [a]	≤ 0,20 % [b]	12.5
Span drift within maintenance interval	≤ 3,0 % [a]	≤ 0,20 % [b]	12.5
Availability	≥ 95,0 %	≥ 98,0 %	12.6
Reproducibility, R_f	≤ 3,3 % [a]	≤ 0,20 % [b]	12.7

a Percentage value as percentage of the upper limit of the certification range.

b Percentage value as oxygen volume concentration (volume fraction).

NOTE The availability during operation is specified e.g. in applicable EU Directives.

8.2.2 AMS for total organic carbon

AMS for measuring total organic carbon shall meet the performance criteria specified in Table 1 and Table 2.

Furthermore, the performance criteria for the effect of oxygen and the response factors specified in Table 3 shall be applied in 10.22.

Table 3 — Performance criteria for AMS measuring Total Organic Carbon (TOC) in laboratory tests

Performance characteristic	Performance criteria
Effect of oxygen	≤ 2,0 % [a]
Range of response factors:	
methane	0,90 to 1,20
aliphatic hydrocarbons	0,90 to 1,10
aromatic hydrocarbons	0,80 to 1,10
dichloromethane	0,75 to 1,15
aliphatic alcohols	0,7 to 1,0
esters and ketones	0,7 to 1,0
organic acids	0,5 to 1,0

a Percentage value as percentage of the upper limit of the certification range.

NOTE 1 EN 12619 and EN 13526 specify performance criteria including response factors for TOC analysers which use flame ionisation detection (FID), particularly when the FID is used as SRM. However, the performance criteria in this European Standard apply to any techniques that can be used to measure TOC or a surrogate for TOC. For example, other techniques, such as Fourier Transform Infrared (FTIR) can be used to measure TOC if AMS using other techniques meet the required performance criteria.

NOTE 2 TOC is measured as volatile organic carbon as defined in EN 12619.

8.3 Particulate matter monitoring AMS

AMS for measuring particulate matter shall meet the performance criteria detailed in Table 4 and Table 5. The maximum allowable deviations (as absolute values) of the measured signals are given as percentages of the upper limit of the certification range.

Table 4 details the performance criteria, which are tested in the laboratory. Table 5 details the performance criteria, which are tested during the 3 month field test.

Table 4 — Performance criteria for particulate matter monitoring AMS in laboratory tests

Performance characteristic	Performance criteria	Test in sub-clause
Response time	≤ 200 s	10.9
Repeatability standard deviation at zero	≤ 2,0 % [a]	10.10
Repeatability standard deviation at zero	≤ 5,0 % [b]	10.10
Lack of fit	≤ 3,0 % [a]	10.12
Zero shift due to ambient temperature change from 20 °C within specified range	≤ 5,0 % [a]	10.14
Span shift due to ambient temperature change from 20 °C within specified range	≤ 5,0% [a]	10.14
Influence of voltage at –15 % and at +10 % from nominal supply voltage	≤ 2,0 % [a]	10.17

[a] Percentage value as percentage of the upper limit of the certification range.

[b] Percentage value as percentage of the emission limit value.

NOTE The response time does not apply to batch-measurement techniques such as beta-ray-attenuation.

Table 5 — Performance criteria for particulate matter monitoring AMS in field tests

Performance characteristic	Performance criteria	Test in sub-clause
Determination coefficient of calibration function, R^2	≥ 0,90	12.1
Response time	≤ 200 s	12.2
Lack of fit	≤ 3,0 %	12.3
Minimum maintenance interval	8 days	12.4
Zero drift within maintenance interval	≤ 3,0 %	12.5
Span drift within maintenance interval	≤ 3,0 %	12.5
Availability	≥ 95,0 %	12.6
Reproducibility, R_{field} — for concentrations > 20 mg/m^3 — for concentrations ≤ 20 mg/m^3	 ≤ 2,0 % ≤ 3,3 %	12.7

Reference materials shall be assessed as appropriate to perform an AST linearity test.

8.4 Flow monitoring AMS

AMS for measuring gas flow shall meet the performance criteria specified in Table 6 and Table 7. The maximum allowable deviations (as absolute values) of the measured signals are given as percentages of the upper limit of the certification range.

Table 6 details the performance criteria, which are tested in the laboratory. Table 7 details the performance criteria, which are tested during the 3 month field test.

Table 6 — Performance criteria for AMS monitoring gas flow in laboratory tests

Performance characteristic	Performance criteria	Test in sub-clause
Response time	≤ 60 s	10.9
Repeatability standard deviation at zero	≤ 2,0 %	10.10
Lack of fit	≤ 3,0 %	10.12
Zero shift due to ambient temperature change from 20 °C within specified range	≤ 5,0 %	10.14
Span shift due to ambient temperature change from 20 °C within specified range	≤ 5,0 %	10.14
Influence of voltage at +15 % and at –10 % from nominal supply voltage	≤ 2,0 %	10.17

Table 7 — Performance criteria for AMS monitoring gas flow in field tests

Performance characteristic	Performance criteria	Test in sub-clause
Determination coefficient of calibration function, R^2	≥ 0,90	12.1
Response time	≤ 60 s	12.2
Minimum maintenance interval	8 days	12.4
Zero drift within maintenance interval	≤ 2,0 %	12.5
Span drift within maintenance interval	≤ 4,0 %	12.5
Availability	≥ 95,0 %	12.6
Reproducibility, R_f	≤ 3,3 %	12.7

9 General test requirements

The test laboratory shall perform all relevant tests on two identical AMS. These two AMS have to be tested in the laboratory and field. Multiple-component AMS shall meet the performance criteria on each individual measured component with all measurement channels operating simultaneously.

NOTE 1 The test is performed in such a way that the material under analysis (measured component) is admitted to all measurement channels in the laboratory test and in the field test.

Changes in the environmental and test conditions shall not have a significant influence on the performance characteristic tested. Therefore, all environmental and test conditions which have an influence on the AMS, shall be kept stable as far as practicable. The environmental and test conditions shall be recorded during the test. All test results shall be reported at standard conditions (0 °C, 1 013 hPa, dry gas).

The test laboratory shall evaluate the performance of the AMS at the lowest certification range possible for the intended application that is chosen by the manufacturer. If the AMS is to be used for industrial plants requiring monitoring over higher measurement ranges, then the test laboratory shall perform selected additional tests to demonstrate satisfactory performance over higher ranges. These additional tests shall at least include evaluations of the response time, lack of fit and cross sensitivity.

NOTE 2 Certification range is selected by the manufacturer in consultation with the test laboratory.

The test requirements specified in Clauses 10 to 13 are the minimum requirements. The tests are divided into two sections, covering general test requirements for all AMS, followed by measured component specific test requirements. They include

— description of the test method,

— evaluation procedure,

— assessment of performance against the relevant performance criterion and

— where appropriate, information on any specialised test equipment.

If a test requires two or more test cycles and the AMS meets the performance criterion by a factor of two or more for the first test then any subsequent testing for this performance characteristic may be omitted.

If a test requires several readings, the average of these readings shall be determined. If a test has to be repeated (several test cycles), the averages of the individual test cycles shall be determined and meet the applicable performance criteria.

The expanded uncertainty of the concentration of test gases at a confidence level of 95 % shall not exceed 3 %. For the lack of fit test, the bench shall provide gases, the concentration of which shall not have an expanded uncertainty greater than 33 % of the lack of fit criterion.

Tests do not have to be performed in the numerical order in this document, as the selection of tests and their order depend on the characteristics and type of individual AMS. However, the first two laboratory tests using test gases are the response time test followed by the lack of fit test.

NOTE 3 The field test is usually carried out after all laboratory tests are passed.

NOTE 4 A short-term drift test performed after the response time test can show that drift is not influencing the results of the other tests. A short-term drift test can be e.g. 24 h long, where a drift at zero and span point of more than 2 % of the upper limit of the certification range indicates that the AMS is not sufficiently stable for the remainder of the tests.

The test laboratory shall document whether the AMS meets all of the relevant performance criteria, and shall record all environmental conditions pertaining during testing.

10 Test procedures for laboratory tests

10.1 AMS for testing

The test laboratory shall check whether the AMS are complete and identical, by examining the appropriate parts specified in the manufacturer's documentation.

The test laboratory shall check that extractive AMS have appropriate provisions for filtration of solids, avoidance of chemical reactions within the sampling system, entrainment effects and effective control of water condensate.

The test laboratory shall include diagrams and photographs of both AMS, in the test report, and copies of the operating manual(s) for the AMS.

NOTE 1 In addition to the analyzer, an AMS can include the sampling probe, the sampling hose, the test gas conditioning facility, any special test components and the operating instructions.

The hardware used shall be photographed and the software version established. Changes in the AMS configuration are not permitted during testing.

NOTE 2 Minor repairs needed to perform the test but without influence on the instrument performance can be carried out, and the test continued.

10.2 CE labelling

If the AMS needs to comply with the requirements for CE labelling as specified in applicable EU Directives, then the test laboratory shall verify whether there is traceable evidence of compliance.

10.3 Security

The AMS shall be set up according to the operating instructions. The test laboratory shall then activate the security mechanisms provided by the AMS manufacturer to prevent inadvertent and unauthorised maladjustment. A check shall then be carried out to establish whether the security mechanisms operate effectively.

NOTE 1 Adjustment can include zero and span adjustments, deletion of data sets, changing averaging times and altering ranges.

NOTE 2 Security mechanisms can include a key or security codes programmed into the AMS which are keyed into the AMS before adjustments are permitted.

10.4 Output ranges and zero point

The test laboratory shall check whether the output ranges on the AMS can be adjusted and whether such ranges are appropriate for the intended applications.

The emission limit values to be monitored with this AMS should be documented, together with an indication of the suitability of the AMS ranges for (i) applicable EU Directives and (ii) other intended applications.

Using reference materials, and by adjusting the zero point on the AMS, the test laboratory shall check that the indicated zero point on the measurement display and output of the AMS is a true living zero, and that the AMS can display both positive and negative readings.

The test laboratory shall use reference materials to verify that the output range is at least twice as great as the certification range.

10.5 Additional data outputs

The test laboratory shall check that the AMS is equipped with an additional data output which allows for example, a recording system or computer to be connected. The test laboratory shall then check that measurement signals displayed on the additional data output are the same results as those on the AMS. The test laboratory shall assess and describe in the test report the mechanism of the additional data output.

10.6 Display of operational status signals

The test laboratory shall assess whether the AMS has a means of displaying and provide data for recording the relevant operational status (e.g. standby, service, malfunction). The test laboratory shall then assess whether each operating mode is correctly identified and outputted by the AMS.

10.7 Prevention or compensation for optical contamination

For optical techniques, the test laboratory shall assess whether contamination of the optical boundary surfaces interferes with the measuring technique.

If contamination interferes with the measuring technique, the effect of contamination on the performance of an optical instrument shall be determined by inserting an optical filter on the process side of the optical surfaces and monitoring the change in signal caused by such contamination. The test should be repeated for both the transmitter and receiver optics and should be performed with an optical filter between 4 % nominal opacity and 10 % nominal opacity. If contamination compensation functions are available on the instruments these should be activated during the tests. For instruments with in-built contamination compensation, the absorption of the optical filter may be specified by the manufacturer to be larger than 10 % in order for the compensation capability of the instrument to be more fully tested. The influence of optical boundary surface soiling on the measurement signal shall be determined while taking into account the physical relationships, and quantified wherever possible through measurements.

There are many types of optical techniques and it is difficult to specify in detail the exact procedures for conducting this test. Therefore the test laboratory should have procedures and adjusting aids, (e.g. reference filter, homogeneous and inhomogeneous particulate coatings) for assessing the effect of soiling on different types of optical system, and then record the types of tests employed in the test report.

The process employed inside the AMS for monitoring the effect of contamination shall be described by the AMS manufacturer in a logical manner. This function shall be operable with the AMS installed and operational. The AMS shall also display when the function is working.

The test report shall contain a description of the AMS-specific method for monitoring soiling. Test results shall be presented in tabular form. The minimum and maximum deviation from rated value shall be documented. The intervals for cleaning the optical boundary surfaces shall be specified for the operating conditions encountered in the performance test.

10.8 Degrees of protection provided by enclosures

The effect of liquid water on the AMS shall be assessed by inspection in relation to EN 60529.

The AMS manufacturer shall provide to the test laboratory the report of testing of the enclosure according to EN 60529. The test laboratory shall assess this test report to ensure compliance with the requirements of 6.8.

10.9 Response time

The test laboratory shall determine the AMS response time using zero and span reference materials (see Figure 1). The zero and span reference materials shall be stable test gases when determining the response time of gas-measuring AMS. The test shall be performed with dry and wet test gases.

NOTE 1 The zero and span material can include surrogates such as filters.

NOTE 2 This test provides the initial stabilisation period, which is then used in other tests described in this European Standard.

NOTE 3 This test can also be combined with the lack of fit test, using the highest concentration in the lack of fit test to determine the response time.

NOTE 4 The response time test is also repeated in the field as real waste gas conditions can influence response times.

The change from zero gas to span gas shall be made almost instantly with use of a suitable valve. The valve outlet shall be mounted directly to the inlet of the sampling system, and both zero gas and span gas shall have the same ”oversupply” which is vented with use of a tee. The gas flows of both zero gas and span gas shall be chosen in such a way that the dead time in the valve and tee can be neglected compared with the lag time of the analyzer.

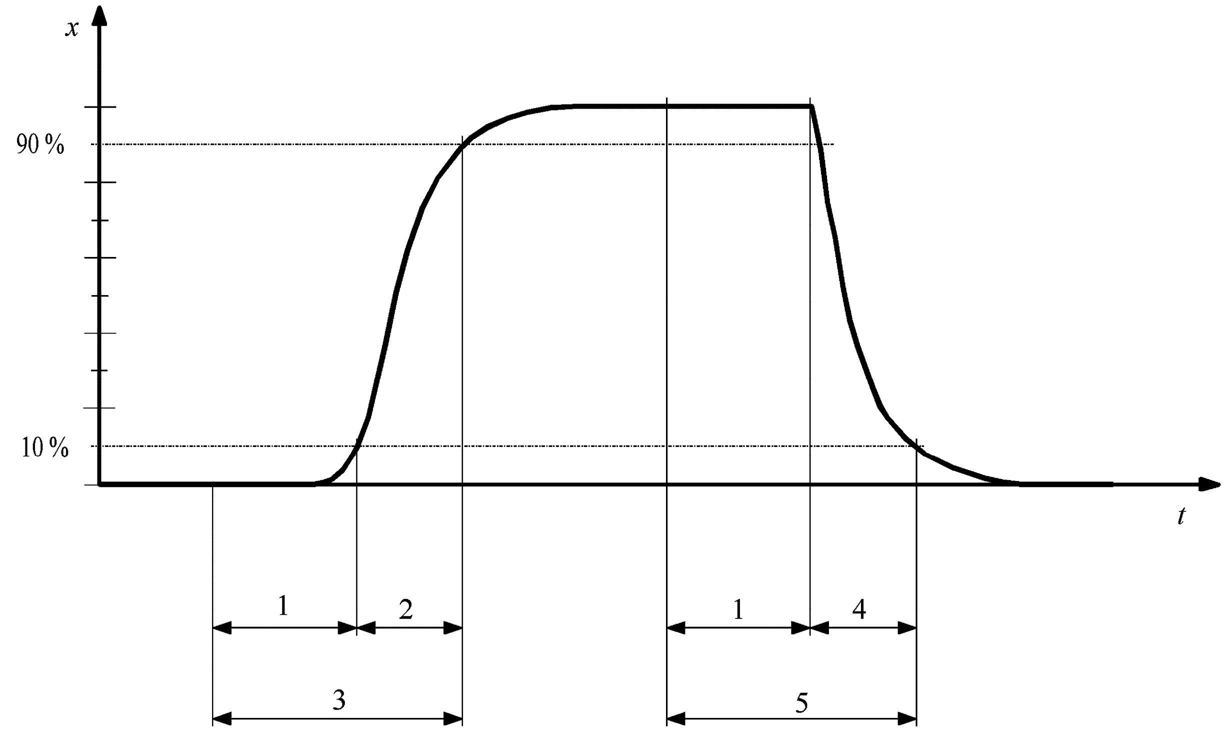
The step change shall be made by switching the valve from zero gas to span gas. This event shall be timed and is the start (t = 0) of the (rise) response time according to Figure 1. When the reading has stabilized, zero gas shall be applied again, and this event is the start (t = 0) of the (fall) response time according to Figure 1. When the reading has stabilized at zero, the whole cycle as shown in Figure 1 is complete.

The elapsed time (response time) between the start of the step change and reaching of 90 % of the difference between the AMS final stable reading of the applied concentration and the AMS reading at the start of the step change shall be determined for both the rise and fall modes.

The whole cycle shall be repeated four times with a time elapse between two experiments of four times the response time but for at least 10 min. If the AMS meets the performance criterion by a factor of two or more for the first test then any subsequent testing may be omitted.

The average of the response times (rise) and the average the response times (fall) shall be calculated.

The larger average value of the response time (rise) and the response time (fall) shall be used as the response time of the AMS and be compared with the applicable performance criteria specified in Clause 8.



Key

1 lag time

2 rise time

3 response time (rise), t_r

4 fall time

5 response time (fall), t_f

x measured signal

t time

Figure 1 — Diagram illustrating the response time

The relative difference in response times shall be calculated according to Equation (1):

$$t_d = \left| \frac{t_r - t_f}{t_r} \right| \tag{1}$$

where

t_d is the relative difference between the response times determined in rise and fall mode;

t_r is the response time (rise);

t_f is the response time (fall).

The values of t_d , t_r and t_f shall be reported individually in the test report.

10.10 Repeatability standard deviation at zero point

The repeatability standard deviation at zero point shall be determined by application of a reference material at the zero point.

If the repeatability standard deviation at zero point is determined during the lack of fit test, the reference material at zero concentration applied during the test shall be used.

The measured signals of the AMS at zero point shall be determined after application of the reference material by waiting the time equivalent to one independent reading and then recording 20 consecutive individual readings.

In the case of oxygen sensors, a zero gas shall have a volume concentration of 0,2 % oxygen. Some oxygen analysers, such as those based on Zirconia sensors, are not suited to responding to pure zero gases. Therefore a lower volume concentration of 2,0 % is to be used.

The measured signals obtained shall be used to determine the repeatability standard deviation at zero point using Equation (2):

$$s_{\mathrm{r}} = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}} \qquad (2)$$

where

s_r is the repeatability standard deviation;

x_i is the ith measured signal;

$\bar{x}$ is the average of the measured signals x_i;

n is the number of measurements, n = 20.

The repeatability standard deviation at zero point shall meet the performance criterion specified in Clause 8.

The individual readings and the repeatability standard deviation at zero point shall be reported.

10.11 Repeatability standard deviation at span point

The repeatability standard deviation at span point shall be determined by application of a reference material at the span point.

If the repeatability standard deviation at span point is determined during the lack of fit test, the highest value of reference material applied during the test shall be used.

The measured signals of the AMS at span point shall be determined after application of the reference material by waiting the time equivalent to one independent reading and then recording 20 consecutive individual readings. The measured signals obtained shall be used to determine the repeatability standard deviation at span using Equation (2).

The repeatability standard deviation at span shall meet the performance criteria specified in Clause 8.

The individual readings and the repeatability standard deviation at span shall be reported.

10.12 Lack of fit

The test laboratory shall perform the lack of fit test according to Annex C. The linearity of the response of the AMS shall be checked using at least seven different reference materials, including a zero concentration.

This test requires the following equipment:

— gas mixing system compliant with national standards and with a maximum expanded uncertainty of 33 % of the lack of fit criterion;

— test standards (e.g. zero gas, test gas of suitable concentration, reference materials);

— data recording system.

The reference material with zero concentration, as well as the reference materials shall have a verifiable quantity and quality.

In case of gaseous reference materials, these reference materials can be obtained from different gas cylinders or can be prepared by means of a calibrated dilution system from one single gas concentration. The test gas concentrations shall be selected so that the measured values are equally spaced over the certification range. It is necessary to know the values of the ratios of their concentrations precisely enough so that an incorrect failure of the lack of fit test does not occur. The dry or wet test gases shall be applied to the inlet of the AMS.

The reference materials shall be applied in an order, which avoids hysteresis effects.

NOTE 1 For the application of seven reference materials, avoidance of hysteresis effects can be achieved e.g. by the following sequence:

— reference material with zero concentration;

— reference material concentration approximately 70 % of the upper limit of the certification range;

— reference material concentration approximately 40 % of the upper limit of the certification range;

— reference material with zero concentration;

— reference material concentration approximately 60 % of the upper limit of the certification range;

— reference material concentration approximately 10 % of the upper limit of the certification range;

— reference material concentration approximately 30 % of the upper limit of the certification range;

— reference material concentration approximately 90 % of the upper limit of the certification range;

— reference material with zero concentration.

After each change in concentration, the measured signals of the AMS shall be determined by waiting the time equivalent to one independent reading and then recording three consecutive individual readings. The three individual readings shall be averaged.

The test shall be repeated three times. If the AMS meets the performance criterion by a factor of two or more for the first test then any subsequent testing may be omitted.

NOTE 2 This procedure means that the quality of the reference material can influence the result of the tests. It should be noted, however, that it is the result that leads to a pass or failure in the test. In some cases, a reference material with a higher quality can change the result from fail to pass.

NOTE 3 Special care should be taken, when handling NH_3, HCl or HF in dry gases. For example, particular surface reactions in tubing can result in very long response time, which is not representative of the response time for humid gases.

NOTE 4 Where no other method is practicable, the linearity test can also be performed with the aid of reference materials such as grating filters or gas filters.

NOTE 5 The reference material at zero concentration applied during the lack of fit test can also be used to determine the repeatability standard deviation at zero point (see 10.10).

NOTE 6 The highest value of reference material applied during the lack of fit test can also be used to determine the repeatability standard deviation at span point (see 10.11).

NOTE 7 This test is performed in the same way for supplementary ranges (see 5.2.2).

The lack of fit shall be evaluated according to Annex C. In this test procedure, a regression line is established between the instrument readings of the AMS (x values) and the reference material values (c values). In the next step, the average of AMS readings at each level is calculated. Then the deviation (residual) of this average to the regression line is calculated.

The relative residuals $d_{c,\mathrm{rel}}$ shall meet the applicable performance criteria specified in Clause 8 for all reference materials applied.

Test records, raw data and results of AMS characteristics shall be documented. The individual values shall be presented together with a statement of the test period for each AMS. The test set-up and ambient conditions shall also be documented.

10.13 Zero and span drift

The aim of this test is to determine that the AMS allows the recording of zero and span drift to satisfy the QAL3 requirements according to EN 14181.

NOTE A short-term drift test performed after the response time test can show that drift is not influencing the results of the other tests. A short-term drift test could be e.g. 24 h long, where a drift at zero and span point of more than 2 % of the upper limit of the certification range indicates that the AMS is not sufficiently stable for the remainder of the tests. Zero and span point drift are checked individually. Span point drift is not corrected for zero point drift.

If the AMS has an automatic mechanism for correcting drift, then the test laboratory shall verify its effectiveness by simulating drift by a procedure to be agreed between the manufacturer and test laboratory. Following any automatic corrections, the test laboratory shall check the zero point and span points using reference materials, and verify the accuracy of these readings.

The test laboratory shall check that the AMS indicates a fault and/or a need for maintenance, if the automatic adjustment is not capable of bringing the AMS within normal operational conditions.

In the test report, the test laboratory shall describe any mechanisms for automatic corrections of drift, the data recorded during the simulated drift checks and the maintenance interval.

If the QAL3 procedure as described in EN 14181 is incorporated within the AMS, this procedure shall be assessed by the test laboratory.

10.14 Influence of ambient temperature

The test laboratory shall determine how the zero and span values of the AMS are influenced by changes in ambient temperature by using a climatic chamber, which can control ambient temperature from –20 °C to +50 °C, within limits of ±1,0 K.

In the case of AMS installed outdoors, the following temperatures shall be set in the climatic chamber in the given order of sequence:

20 °C → 0 °C → –20 °C → 20 °C → 50 °C → 20 °C.

In the case of AMS installed at temperature-controlled locations, the following temperatures shall be set in the given order of sequence:

20 °C → 5 °C → 20 °C → 40 °C → 20 °C.

After a sufficient equilibration period, the measured signals of the AMS at zero point and at span point shall be determined at each temperature by waiting the time equivalent to one independent reading and then recording three consecutive individual readings. The three individual readings shall be averaged.

The test laboratory shall wait at least 6 h between each temperature change in the environmental chamber, to allow the AMS to equilibrate, before taking further readings.

Alternatively, the test laboratory may monitor the reading from the AMS following each temperature change. If the instrument stabilises in less than six hours, then the test laboratory may reduce the equilibration period. However, the test laboratory shall record objective and verifiable evidence to support this.

The AMS shall remain switched on when varying the ambient temperature in the environmental chamber.

The deviations between the average reading at each temperature and the average reading at 20 °C shall be determined. The deviations shall meet the applicable performance criteria specified in Clause 8 for all temperatures.

The test shall be repeated three times at the zero point and three times at the span point. If the AMS meets the performance criterion by a factor of two or more for the first test then any subsequent testing may be omitted.

The individual readings, averages and deviations at each temperature as well as the maximum deviation at zero point and at span point shall be reported.

In addition, the test laboratory shall determine and report the maximum sensitivity coefficient for the temperature dependence. The sensitivity coefficients at each temperature shall be calculated by Equation (3):

$$b_\mathrm{t} = \frac{(x_i - x_{i-1})}{(T_i - T_{i-1})} \tag{3}$$

where

b_t is the sensitivity coefficient of ambient temperature;

x_i is the average reading at temperature T_i;

x_{i-1} is the average reading at temperature T_{i-1};

T_i is the current temperature in the test cycle;

T_{i-1} is the previous temperature in the test cycle.

NOTE A graph showing the results of the examination can be provided in the report.

10.15 Influence of sample gas pressure

The test laboratory shall determine the influence of variations in sample gas pressure on the response of the AMS. The sample shall be nitrogen containing the measured component at a concentration of between 70 % and 90 % of the upper limit of the certification range.

NOTE The effect of sample gas pressure typically applies to in-situ AMS, but not to extractive AMS, since the sample gas is conditioned and typically not subject to significant variations of temperature and pressure once within the analyser.

The test laboratory shall measure the output signal of the AMS when the sample gas pressure is at the following levels:

a) ambient atmospheric pressure;

b) approximately 3 kPa above ambient atmospheric pressure, within limits of ±0,2 kPa;

c) approximately 3 kPa below ambient atmospheric pressure, within limits of ±0,2 kPa.

During the measurement period the temperature shall be held stable to within ±1,0 K.

The measured signals of the AMS shall be determined at each pressure by waiting the time equivalent to one independent reading and then recording three consecutive individual readings. The three individual readings shall be averaged.

The deviations between the average reading at each pressure and the average reading at the ambient atmospheric pressure shall be determined. The deviations shall meet the applicable performance criteria specified in Clause 8.

The individual readings, averages and deviations at each pressure as well as the maximum deviation shall be reported.

In addition, the test laboratory shall determine and report the sensitivity coefficient for the pressure dependence. The sensitivity coefficient shall be calculated by Equation (4):

$$b_{\mathrm{p}} = \frac{(x_2 - x_1)}{(p_2 - p_1)} \tag{4}$$

where

b_{p} is the sensitivity coefficient of sample gas pressure;

x_1 is the average reading at sample gas pressure p_1;

x_2 is the average reading at sample gas pressure p_2;

p_1 is the lower sample gas pressure;

p_2 is the higher sample gas pressure.

10.16 Influence of the sample gas flow for extractive AMS

The AMS shall initially be operated with the flow rate prescribed by the manufacturer. This flow rate shall then be changed to the lowest flow rate specified by the manufacturer.

NOTE Influence of the sample gas flow typically applies to extractive AMS, since in-situ AMS mostly are not influenced by flow rate.

If the manufacturer's documentation permits only minor tolerances these are binding and shall not be extended.

The measured signals of the AMS at the zero point and span point shall be determined at both flow rates by waiting the time equivalent to one independent reading and then recording three consecutive individual readings. The three individual readings shall be averaged.

The deviation between the average readings at both flow rates shall be determined. The deviation shall meet the applicable performance criteria specified in Clause 8.

This test shall be repeated three times at the zero point and three times at the span point. If the AMS meets the performance criterion by a factor of two or more for the first test then any subsequent testing may be omitted.

The individual readings, averages and the deviations as well as the maximum deviation shall be reported.

The functionality of the status signal shall be tested at the same time.

In addition, the test laboratory shall determine and report the sensitivity coefficient for the flow rate dependence. The sensitivity coefficient shall be calculated by Equation (5):

$$b_\mathrm{f} = \frac{(x_2 - x_1)}{(r_2 - r_1)} \tag{5}$$

where

b_f is the sensitivity coefficient of flow rate;

x_1 is the average reading at flow rate r_1;

x_2 is the average reading at flow rate r_2;

r_1 is the nominal flow rate;

r_2 is the lowest specified flow rate.

10.17 Influence of voltage variations

The supply voltage to the AMS shall be varied, using an isolating transformer, in steps of 5 % from the nominal supply voltage to at least the upper and the lower limits specified in Clause 8.

The measured signals of the AMS at zero point and at span point shall be determined at each voltage by waiting the time equivalent to one independent reading and then recording three consecutive individual readings. The three individual readings shall be averaged.

NOTE After changes in voltage the AMS can need time to stabilize.

The deviations between the average readings at each voltage and the average reading at the nominal supply voltage shall be determined. The deviations shall meet the applicable performance criteria specified in Clause 8 for all voltages.

This test shall be repeated three times at the zero point and three times at the span point. If the AMS meets the performance criterion by a factor of two or more for the first test then any subsequent testing may be omitted.

The individual readings, averages and deviations at each voltage as well as the maximum deviation at zero point and at span point shall be reported.

In addition, the test laboratory shall determine and report the sensitivity coefficient for the voltage dependence. The sensitivity coefficient shall be calculated by Equation (6):

$$b_\mathrm{v} = \frac{(x_2 - x_1)}{(U_2 - U_1)} \tag{6}$$

where

b_v is the sensitivity coefficient of supply voltage;

x_1 is the average reading at voltage U_1;

x_2 is the average reading at voltage U_2;

U_1 is the minimum voltage specified by the manufacturer;

U_2 is the maximum voltage specified by the manufacturer.

For reporting the voltage dependence, the highest value of the results at zero and span point shall be taken.

10.18 Influence of vibration

The AMS shall be examined in the laboratory and in the field in respect to whether normal vibrations affect the performance of the AMS, if the conditions of use specified by the manufacturer demand that a vibration test be performed. In this case, the measured signals of the AMS at zero point and at span point shall be determined before and after the vibration test by waiting the time equivalent to one independent reading and then recording three consecutive individual readings. The three individual readings shall be averaged.

The vibration test, if required, shall be applied to duct-mounted parts of the AMS only and shall be made with reference to EN 60068-1 and the appropriate sections of EN 60068-2. The instrument shall be subjected to vibration on three perpendicular axes in turn, with a swept range of frequencies from 10 Hz to 160 Hz at one octave per minute and at a vibration acceleration chosen by the manufacturer and assessed by the test laboratory to be proper for the application that the AMS is intended for. If any resonant frequencies are observed, a vibration test shall be carried out at each observed frequency for a period of 2 min.

This shall be followed by a functional test. If no resonant frequencies are observed, a vibration test shall be made at a frequency of 25 Hz for a period of 2 min. This shall be followed by a functional test. Any influence on the AMS from vibrations at the site of installation shall be assessed. Remedial measures having proved necessary in field testing shall be described.

The deviations between the average readings before and after the vibration tests shall be determined. All deviations shall meet the applicable performance criteria specified in Clause 8.

The individual readings, averages and deviations at each vibration test as well as the maximum deviation shall be reported.

10.19 Cross-sensitivity

The influence of potentially interfering substances also present in the waste gas shall be determined by admitting test gas mixtures to the input of the complete AMS (upstream of the test gas cooler, where present). The gas mixtures shall be produced with a mixing system in which an interferent is added to the gases for zero point and span point. The mixing system shall be compliant with national standards and shall have a maximum expanded uncertainty of 1 %. Reference materials (e.g. gases) shall be certified (traceable to national standards) and shall have an expanded uncertainty no greater than 2 %.

Interferents and their concentrations are defined in relation to the measuring principle and the intended measurement objective. The interferents listed in Annex B shall be examined. The interferents shall be admitted individually.

Test gas without interferent and then with the interferent shall be applied. The measured signals of the AMS shall be determined for each test gas by waiting the time equivalent to one independent reading and then recording three consecutive individual readings. The three individual readings shall be averaged.

The deviations between the average readings with and the average reading without the interferent present at the zero point and span point shall be determined for each interferent.

All positive deviations above 0,5 % of the span gas concentration shall be summed and all negative deviations below –0,5 % of the span gas concentration shall be summed at both the zero point and span point. The maximum of the absolute values of the four summations shall meet the performance criteria specified in Clause 8.

For total organic carbon AMS, the influence of oxygen shall be additionally examined at zero point and span point for an oxygen volume concentration of 3 % and 21 %. Propane should be used as a span gas. The deviation between the average readings at the zero point obtained with an oxygen volume concentration of 3 % and 21 %, and the deviation between the average readings at the span point obtained with an oxygen volume concentration of 3 % and 21 % shall be determined. Both deviations shall meet the performance criteria specified in Clause 8.

The individual readings, averages and deviations at zero point and span point and for all interferents as well as the maximum deviation shall be reported.

10.20 Excursion of measurement beam of cross-stack in-situ AMS

The test laboratory shall gradually and precisely deflect the transmitter and receiver assemblies of the AMS in the horizontal and vertical planes, and then record the measured signals using reference materials.

NOTE 1 This test typically applies to cross-stack in-situ optical techniques. The test also applies to extractive AMS with separate transmitter and receiver assemblies.

NOTE 2 This testing requires calibration standards (e.g. reference filters) and an optical bench.

NOTE 3 Typically the experimental path length for this test is 2 m to 3 m, although the test should be performed at the maximum path length practical.

Deflections shall be carried out for both the position of the zero point and as well as for that of a span point for approximately 70 % to 90 % of the upper limit of the certification range over two typical measurement path lengths. The deflection is to be performed in incremental steps of approximately 0,05° in the angle range demanded.

The range of deflection shall be equal to at least twice the angle specified by the manufacturer. It should also be tested as far as the deflection limit permitted by the assemblies - if necessary in larger increments.

The efficiency of any manual optical adjustment facilities shall be examined at least in qualitative terms. Automatic adjustment processes shall be activated and included in the test.

The measured signals obtained for the various test steps shall be included in tabular form in the test report. These measured signals shall be paired up with the deflection angles.

The maximum permissible deflection angles shall be stated within which the AMS satisfies the performance criterion. In the case of automatically aligning AMS, the manner of operation shall be described and verified by means of test results.

10.21 Converter efficiency for NO_x AMS

The test laboratory shall determine the efficiency of the NO_x converter before and after the field test by carrying out the following procedure.

The following equipment is required:

— source of nitrogen monoxide such as a compressed gas cylinder containing nitrogen monoxide in nitrogen at a concentration of the order of 80 % of the upper limit of the certification range; the actual concentration need not be known provided that it remains constant throughout the test;

— source of oxygen, such as a compressed gas cylinder containing air or oxygen;

— an ozone generator, capable of providing varying amounts of ozone from oxygen.

The test laboratory shall ensure that the total flow rate of nitrogen monoxide and air (or oxygen) is greater than the flow rate of gas through the analyser. In all of the following steps, determine the responses of the analyser to both the nitrogen monoxide and total NO_x.

NOTE 1 This procedure evaluates the concentration of nitrogen dioxide being produced, which should be in the range 10 % to 90 % of the NO_x.

Turn off the ozone generator. Then the concentration $c_{\mathrm{NOx},0}$ of total NO_x and the concentration $c_{\mathrm{NO},0}$ of NO shall be determined by waiting the time equivalent to one independent reading, then recording three consecutive individual readings and averaging the three individual readings for each concentration.

Turn on the ozone generator and vary the output of the ozone generator to obtain at least five different ozone concentrations. Then the displayed concentrations of total NO_x ($c_{\mathrm{NOx},1}$, $c_{\mathrm{NOx},2}$, …, $c_{\mathrm{NOx},n}$) and nitrogen monoxide ($c_{\mathrm{NO},1}$, $c_{\mathrm{NO},2}$, …, $c_{\mathrm{NO},n}$) shall be determined by waiting the time equivalent to one independent reading, then recording three consecutive individual readings and averaging the three individual readings for each concentration.

NOTE 2 Ozone is formed which reacts with the nitrogen monoxide to produce nitrogen dioxide before the gases enter the analyser

Calculate the ratios $c_{\mathrm{NOx},1}/c_{\mathrm{NOx},0}$, $c_{\mathrm{NOx},2}/c_{\mathrm{NOx},0}$, ... , $c_{\mathrm{NOx},n}/c_{\mathrm{NOx},0}$, which should be as close to unity as possible. Therefore, the concentration of total NO_x should be constant in each case and independent of the ratio of the concentrations of nitrogen dioxide to nitrogen monoxide.

Calculate the converter efficiency E_i for each ozone concentration, expressed as a percentage, using Equation (7):

$$E_i = \frac{(c_{\mathrm{NOx},i} - c_{\mathrm{NO},i}) - (c_{\mathrm{NOx},0} - c_{\mathrm{NO},0})}{c_{\mathrm{NO},0} - c_{\mathrm{NO},i}} \times 100\ \% \tag{7}$$

where

E_i is the converter efficiency at setting i of the ozone generator;

$c_{\mathrm{NOx},0}$ is the concentration of total NO_x with ozone generator switched-off;

$c_{\mathrm{NO},0}$ is the concentration of NO with ozone generator switched-off;

$c_{\mathrm{NOx},i}$ is the concentration of total NO_x with ozone generator at setting i (i = 1 to n);

$c_{\mathrm{NO},i}$ is the concentration of NO with ozone generator at setting i (i = 1 to n).

All converter efficiencies shall meet the performance criterion specified in Clause 8.

The results shall be presented in tabular form.

NOTE 3 Determination of converter efficiency with other methods is acceptable for converter test during AST specified in EN 14181, if these methods are well described in the AST report.

10.22 Response factors

The response factors (weighting factors) for total organic carbon measuring AMS shall be evaluated using defined test gas concentrations admitted to the AMS from test gas containers or by evaporating produced mixtures.

This test requires certified reference materials and a gas-mixing system compliant with national standards and with an expanded uncertainty no greater than 2 % of the combined concentration.

The evaluation shall cover at least the following organic compounds, as specified in EN 12619 and EN 13526:

- methane;
- ethane;
- benzene;
- toluene;
- dichloromethane;
- test gas mixture according to EN 12619 (methane, ethane, toluene, benzene, dichloromethane, oxygen, carbon dioxide, carbon monoxide, nitrogen).

For total organic carbon measuring AMS for use in measuring emissions from waste incinerators, the response factors of the following organic compounds shall also be evaluated:

- propane;
- ethyne;
- ethyl benzene;
- p-xylene;
- chlorobenzene;
- tetrachloroethylene;
- n-butane;
- n-hexane;
- n-octane;
- isooctane;
- propene;
- methanol;
- butanol;
- acetic acid;
- acetic acid methyl ester;
- trichloromethane;
- trichloroethylene.

The measured signals of the AMS shall be determined for propane and for each organic compound by waiting the time equivalent to one independent reading and then recording three consecutive individual readings. The three individual readings shall be averaged for propane and for each organic compound.

The response factors shall be calculated for each organic compound in accordance with EN 12619 by the Equation (8):

$$f_i = \frac{\left(\frac{x_i}{c_i}\right)}{\left(\frac{x_{\mathrm{ref}}}{c_{\mathrm{ref}}}\right)} \tag{8}$$

where

f_i is the carbon-related response factor for substance i;

x_i is the average of the measured signals for substance i;

x_{ref} is the average of the measured signals for propane;

c_i is the carbon mass concentration of substance i at 273 K and 1 013 hPa;

c_{ref} is the carbon mass concentration of propane at 273 K and 1 013 hPa.

The response factors determined shall meet the applicable performance criteria specified in Table 3.

The individual readings, averages and response factors for each organic compound shall be reported.

11 Requirements for the field test

11.1 Provisions

The field test is an endurance test on an industrial facility appropriate to the AMS's field of application. When selecting the industrial facility, the test laboratory shall ensure that the mass concentrations of the measured component are available in a concentration sufficient to assess the measurement results. This is usually the case if the mass concentration is in the range of 30 % to about 100 % of the daily emission limit value being monitored.

The measurement site shall be selected in accordance with EN 15259. The field test shall be performed with two complete and identical AMS under repeatability conditions. Any additional equipment required is specified in the various test procedures.

In the case of in-situ measuring methods, the two AMS shall be arranged in such a way that a representative measurement is ensured for both AMS in the same measurement cross section.

In exceptional cases, which shall be justified, a common sampling system may be used for both AMS under test.

NOTE For example, in the field test on a plant exhibiting only slight variation from the pollutant concentration being measured and therefore requiring the addition, i.e. input of high concentrations, of the measured component into the sample gas flow.

11.2 Field test duration

The field test duration shall be at least three months and shall not be interrupted. Only in exceptional cases, which shall be justified (e.g. in the case of operation-related interruptions or change of site), shorter testing periods may be included in the field test. The total duration of the shorter testing periods shall be at least three months. During the field test, the performance criteria shall be determined under near-practice and realistic conditions.

The measured signals from both AMS throughout the field test shall be recorded as 1 min values. Depending on the task, internal test cycles shall be separately recorded and assessed.

The test laboratory shall document and maintain records of all data from the field test.

12 Test procedures common to all AMS for field tests

12.1 Calibration function

The test laboratory shall determine the calibration function for the individual measured components of the AMS during the field test, by performing parallel measurements with the SRM. The calibration function shall be determined on the basis of at least 15 measurements in accordance with EN 14181.

The calibration function shall be determined twice, once at the beginning and once again at the end of the field test. Both calibration functions shall meet the performance criteria for measurement uncertainty specified in the applicable regulations. The peripheral parameters (e.g. moisture content, temperature and oxygen content) used to standardise the measured results should be the same for AMS and SRM measured values to calculate the variability only for the tested AMS without influence of peripheral parameters. Furthermore, the validity of the second calibration function shall be assessed according to Equation (18) of EN 14181:2004 by use of all results of the parallel measurements.

The sampling point for the AMS shall be selected in accordance with EN 15259.

NOTE Compliance with EN 15259 for the selection of the AMS sampling point eliminates or at least minimises any systematic deviation caused by spatial and temporary inhomogenieties in the mass concentration or volumetric concentration of individual measured components.

In the case of particulate matter monitoring AMS using new measurement principles, which have not been certified under the conditions of this standard before, basic calibration shall be conducted in a wind tunnel prior to field testing. This basic calibration shall show that the AMS can be calibrated under ideal conditions and that the measurement process is usable in practice.

The calibration function during basic calibration can be determined by admitting a known quality and quantity of test dust to the wind tunnel. Several test dust concentrations evenly distributed across the measuring range are selected for this purpose. The comparative measurements in the wind tunnel shall be conducted in accordance with EN 13284-1 and EN 13284-2.

If any effects of waste gas temperature and pressure are observed during the field tests, the test laboratory shall note these effects in the test report.

12.2 Response time

The response time shall be evaluated by admitting reference material at the zero and span points to the input of the complete AMS. The test shall be performed at least two times, once at the beginning and once at the end of the field test, using the procedures described in 10.9.

NOTE This test can be combined with the test for lack of fit, using the procedures described in 10.9 and 10.12.

12.3 Lack of fit

The lack of fit shall be determined at least twice during the field test, using the procedures described in 10.12. This is done to provide information for the AST as required by EN 14181. The residuals of the average concentration at each concentration level to the regression line relative to the upper limit of the certification range shall meet the performance criterion specified in Clause 8.

12.4 Maintenance interval

The test laboratory shall determine the maintenance work that is necessary for the AMS to work properly as well as the intervals at which such maintenance work shall be performed. The recommendations of the instrument manufacturer should be taken into account.

The maintenance interval shall be derived from the shortest interval between the requisite maintenance work operations. This also includes manual zero and span point checks.

If the AMS does not require any service, the maintenance interval is determined by the drift behaviour. In order to determine drift behaviour, the AMS shall be adjusted at the start of testing with the relevant test gases and test filters. The zero and span points shall be checked at regular intervals (e.g. once a week) during the further course of testing. The maintenance interval shall be defined as the time period between the start of the test and the last time when the deviation remained within the permissible drift.

The maximum allowable maintenance interval for a three-month field test shall be one month. Extending the maintenance interval to one year necessitates long-term studies as specified in Table 8. The test laboratory has to describe the minimum amount of maintenance work to be performed within the maintenance interval.

Table 8 — Maximum allowable maintenance intervals

Field test duration	Maximum allowable maintenance interval
3 months	1 month
6 months	3 month
12 months	6 month
24 months	12 month

12.5 Zero and span drift

The test shall be performed using two AMS of identical design as part of the field test in the form of paired measurements in the smallest measuring range tested. The position of the zero point and of the span point shall be determined ten times manually on a pollutant-free measurement section, if necessary using the test standard provided with the AMS, at intervals of a maximum of four weeks over the period of the field test. A different interval may also be selected in substantiated cases.

In the case of AMS with automatic recording of zero point and span point drift, the display readings shall also be acquired and recorded over the period intended for the maintenance interval as well as over the period of the field test.

Manual re-adjustment to the AMS characteristic at the zero point or at the span point shall only be permissible if the test laboratory establishes that the permissible level of drift is exceeded during an inspection interval. The maintenance work operations defined by the manufacturer of the AMS shall be performed at the prescribed intervals and included in the test.

All manually determined zero point and span point values are used for the purpose of evaluation and presented in tabular form together with the associated times. AMS-internal control values issued automatically by the AMS as signal values are checked for adherence to the permissible levels of drift.

The time between two test intervals in which drift is exceeded shall, if necessary, be established. This is based on the levels of drift permitted under current performance criteria.

Both the manually determined zero point and span point values as well as any control values displayed automatically by the AMS shall be assessed. The minimum and maximum deviation from rated value shall be documented.

In the case of AMS with automatic zero-point and reference-point correction facility, the maximum technically permissible amount of correction shall be specified or determined from the test results. The levels of drift for the zero point shall be related to the relevant measuring range and those for the span point to the rated value.

Drift of zero point and span point during maintenance interval shall be recorded for calculation of the influence on measurement uncertainty.

12.6 Availability

The test laboratory shall determine the availability of the AMS by recording the duration of the field test and all interruptions to the normal monitoring functionality of the AMS.

NOTE Interruptions include malfunctions, servicing work, zero and span point checks.

The availability V in percent shall be determined using Equation (9) with the aid of the total operating time t_{tot} and the outage time t_o:

$$V = \frac{t_{tot} - t_o}{t_{tot}} \times 100\ \% \tag{9}$$

The results shall be summarised in tabular form. Table 9 provides an example.

Table 9 — Summary of availability test results

		AMS 1	AMS 2
Total operating time t_{tot}	h		
Outage time t_o – AMS internal setting times – AMS malfunction and repairs – Maintenance, adjustment	 h h h		
Availability V	%		

The test records with the raw data and results shall be documented.

12.7 Reproducibility

Reproducibility shall be determined during the three month field test from simultaneous, continuous measurements by means of two identical AMS at the same measurement point (paired measurements) and an electronic data recording system with a memory capacity of at least four weeks and a sampling rate of at least four times during the AMS averaging period.

The test shall be carried out in the smallest measuring range under test. When selecting the plant, it is necessary to ensure that in the range of 30 % to about 100 % of the monitored ELV the mass concentrations of the measured component are available in a concentration sufficient to assess the measured results.

The measured signals of both AMS (raw values as analogue or digital output signals without any conversion) shall be recorded as individual values (e.g. minute mean values) on an electronic data register. The relevant status signals, such as measurement, malfunction and maintenance, shall also be recorded. Taking into account status signals, the individual values shall be condensed into 30 min mean values, provided that for each 30 min at a minimum 20 min are covered by individual values. Measured signals from malfunction, maintenance or test cycles taking place in the AMS shall not be taken into consideration for evaluation.

In specific cases, shorter averaging time of measured value pairs, e.g. 10 min, may be used, if the measured component has to be evaluated on this averaging time (e.g. for CO), or if higher concentrations of the measured component are not available over prolonged intervals as a result of the dynamics of the emission profile.

At the end of the field test, the reproducibility shall be calculated on the basis of all valid paired values, i.e. the condensed measured signals from the AMS, accrued throughout the entire period of the field test with both AMS in accordance with Equation (10) using the standard deviation of the difference of the paired measurements as given by Equation (11) and with a statistical confidence of 95 % for the t-distribution (two-sided):

$$R_\mathrm{f} = t_{n-1;\,0{,}95} \times s_\mathrm{D} \tag{10}$$

$$s_\mathrm{D} = \sqrt{\frac{\sum_{i=1}^{n}\left(x_{1,i} - x_{2,i}\right)^2}{2n}} \tag{11}$$

where

R_f is the reproducibility under field conditions;

$t_{n-1;\,0{,}95}$ is the two-sided Students t-factor at a confidence level of 95 % with a number of degrees of freedom n–1;

s_D is the standard deviation from paired measurements;

$x_{1,\,i}$ is the ith measured signal of the first measuring system;

$x_{2,\,i}$ is the ith measured signal of the second measuring system;

n is the number of parallel measurements.

NOTE The determination of the reproducibility under field conditions is in accordance with ISO 5725-2.

In the test report, the paired values shall be plotted on a graph in accordance with Figure 2. In the graph, the measured value pairs below 30 % of the ELV shall be identified separately.

The reproducibility calculation results shall deliver the following information in tabular form:

— concentration range of value pairs (e.g. in mg/m^3);

— number of value pairs above 30 % of the ELV;

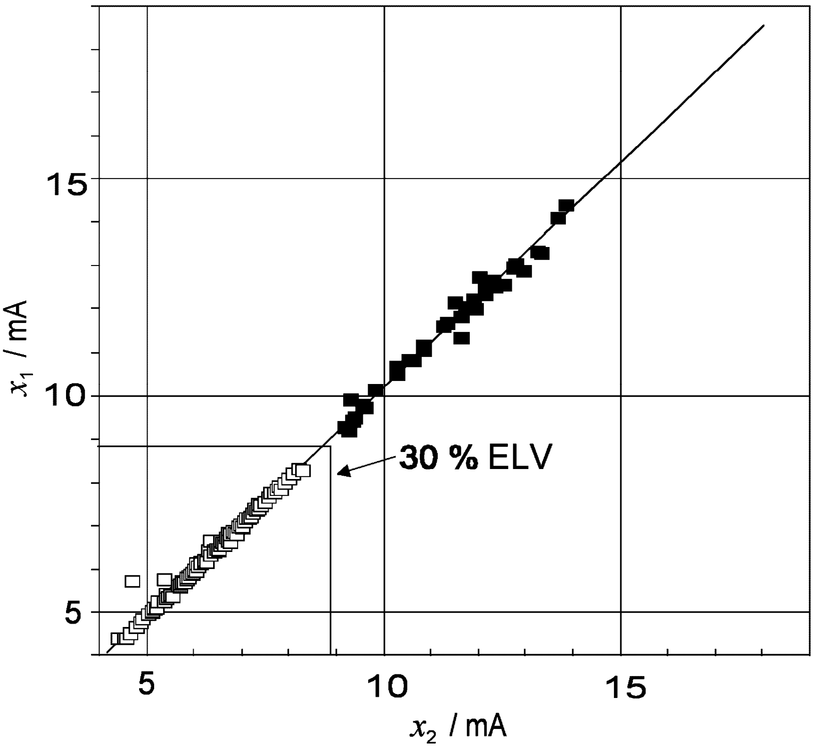
— number of value pairs below 30 % of the ELV;

— total number of valid value pairs in the test period;

— reproducibility under field conditions, related to the measuring signal range (e.g. 4 mA to 20 mA for analogue outputs);

— measuring range (e.g. signal range: 4 mA to 20 mA, concentration range: 0 mg/m^3 to 15 mg/m^3).

For calculating the uncertainty, the standard deviation from the paired measurement values shall be documented.



Keys

–	regression line	$x_{1,i}$, $x_{2,i}$	measured signals from the first and second AMS
	measured signals not used for evaluation	ELV	emission limit value
■	measured signals used for evaluation		

Figure 2 — Graphical representation of measured signals used for determination of reproducibility under field conditions

12.8 Contamination check of in-situ systems

To evaluate the likelihood of contamination problems occurring in plant conditions and the performance of any contamination compensation features in the field test, the test laboratory shall determine the functionality and reliability of anti-contamination and de-contamination functions during the field test by regularly checking the state of contamination of the optical boundary surfaces. Any relationship between the degree of soiling and the resultant measurement error should be quantified.

The response of the AMS to soiling shall be determined in the field test by means of visual checks and, for example, by determining deviation from the rated values of the AMS characteristic.

If required, the AMS shall be provided with recommended air purging systems for 3 months as part of the field test. At the end of the test, the effect of contamination shall be evaluated by removing the sensor from the duct and monitoring the instrument output with an external span material being inserted in the measuring volume. The optical surfaces shall then be cleaned. After self-check and correction initiated automatically or manually by the test laboratory, the output signal has to be checked with the same span material. The results with clean and soiled optical surfaces shall differ by no more than 2,0 % of the upper limit of the certification range.

The test report shall contain a description of the AMS-specific method for monitoring soiling. Test results shall be presented in tabular form. The minimum and maximum deviation from rated value shall be documented. The intervals for cleaning the optical boundary surfaces shall be specified for the operating conditions encountered in the suitability test.

13 Test procedures for particulate AMS

13.1 Lack of fit

The test laboratory shall determine the lack of fit of particulate AMS using appropriate reference materials. These reference materials shall be provided by the AMS manufacturer. The test laboratory shall assess that the chosen reference material applied to the AMS as an independent check of the instrument's operation is capable of monitoring any credible change in instrument response, not caused by changes in the measured component or stack condition.

NOTE The linearity of the characteristic of AMS for particulate emissions can in general only be tested with test standards that generate a suitable measured signal in the output range.

The test laboratory shall determine the lack of fit with a least four values including zero distributed evenly over the measuring range. In the case of flow sampling methods, comparable reference materials shall be used that are appropriate to the particular measuring technique concerned.

The lack of fit test shall be performed at the start and at the end of the field test. If test signals are unstable in relation to time, the respective measured value shall be tested for constancy with the aid of the data recording system.

The test results, together with all individual values and computed deviations, shall be presented in tabular form in the concluding test report. Presentation of results in graph form is normally not necessary on account of the few graduations available over the measuring range.

The values shall normally be shown as the raw analogue or digital output signals from the AMS.

13.2 Extractive AMS

In case of extractive AMS for particulate matter with isokinetic sampling, the extraction shall be in accordance with the requirements of EN 13284-1. In case of non-isokinetic extraction the test laboratory shall determine and assess this influence at a plant with variations in the flow or in a wind tunnel.

For AMS measuring the mass directly, the extraction flow rate shall be measured within an uncertainty of 2,0 %.

14 Measurement uncertainty

The values of the uncertainties determined during the field and laboratory test shall be used to determine the combined standard uncertainty of the AMS measured values according to EN ISO 14956. When calculating the combined standard uncertainty, the repeatability in the laboratory or the reproducibility in the field shall be used, whichever is greater.

NOTE In addition to the uncertainty contribution from the tested AMS, EN ISO 14956 also allows to take into account the uncertainty of external input quantities (e.g. measured values from the peripheral instruments on the site used for conversion to standard conditions) when the combined standard uncertainty is calculated according to Equation (4) of EN ISO 14956.

Annex D provides a method of the uncertainty determination.

The total uncertainty of the AMS determined from the tests according this standard should be at least 25 % below the maximum permissible uncertainty specified e.g. in applicable regulations. A sufficient margin for the uncertainty contribution from the individual installation of the AMS is necessary to pass QAL2 and QAL3 of EN 14181 successfully.

The test laboratory shall report the total uncertainty in relation to the maximum permissible uncertainty specified e.g. in applicable regulations for the intended application.

15 Test report

The test report shall provide a comprehensive and detailed account of the testing and AMS performance.

Annex E specifies requirements for a test report.

NOTE The test report is part of the documentation of the certified AMS.

Annex A
(informative)

Standard reference methods

Table A.1 — Standard reference methods

Measured component	**Standard**	**Issued**
Particulate matter	EN 13284-1	2001
Mercury	EN 13211	2001
Inorganic sulphur compounds		
Sulphur dioxide	EN 14791	2005
Hydrogen sulphide	Applicable national standards	
Inorganic nitrogen compounds		
Nitrogen monoxide	EN 14792	2005
Carbon monoxide	EN 15058	2006
Inorganic chorine compounds		
Hydrogen chloride	EN 1911 (all parts)	1998
Chlorine	Applicable national standards	
Fluorine compounds	Applicable national standards	
Hydrofluoric acid	ISO 15713	2006
Organic measured components		
Hydrocarbons (general)	Applicable national standards	
Hydrocarbons (FID)	EN 12619	1999
	EN 13526	2001
GC determination of organic compounds	EN 13649	2001
Aliphatic aldehydes C1 to C3	Applicable national standards	
Acrylonitrile	Applicable national standards	
1,3-butadiene	Applicable national standards	
PAH	ISO 11338-1, ISO 11338-2	2003
Vinyl chloride	Applicable national standards	
Odours/olfactometry		
Odorous substance concentration	EN 13725	2003
General standards		
Calibration	EN 14181	2004
Measurement planning	EN 15259	2007
Measuring emissions	CEN/TS 15675	2007

Annex B
(normative)

Interferents

Table B.1 — Concentrations of interferents used during cross sensitivity tests

Interferent	Mass or volume concentration	
	Value	Unit
O_2	3 [a] and 21	%
H_2O	30	%
CO	300	mg/m^3
CO_2	15	%
CH_4	50	mg/m^3
N_2O	20	mg/m^3
N_2O (fluidised-bed firing)	100	mg/m^3
NO	300	mg/m^3
NO_2	30	mg/m^3
NH_3	20	mg/m^3
SO2	200	mg/m^3
SO_2 (coal-fired power stations without desulphurisation)	1 000	mg/m^3
HCl	50	mg/m^3
HCl (coal-fired power stations)	200	mg/m^3

[a] A test with 3 % oxygen volume concentration is used instead of a test without interferent.

Annex C
(normative)

Test of linearity

C.1 Description of the test procedure

In this test procedure, a regression line is established between the instrument readings of the AMS (x values) and the reference material values (c values). In the next step, the average of AMS readings at each reference material level is calculated. Then the deviation (residual) of this average to the regression line is calculated.

C.2 Establishment of the regression line

A linear regression for the function in Equation (C.1) is established:

$$x_i = a + B\left(c_i - \bar{c}\right) \tag{C.1}$$

For the calculation, all measurement points are taken into account. The total number of measurement points (n) is equal to the number of reference material levels (including zero) times the number of repetitions (these are the results of the at least three readings) at a particular reference material level.

The coefficient a is obtained by Equation (C.2):

$$a = \frac{1}{n}\sum_{i=1}^{n} x_i \tag{C.2}$$

where

a is the average value of the x values, i. e. the average of the AMS readings;

x_i is the individual AMS reading;

n is the number of measured signals.

The coefficient B is obtained by Equation (C.3):

$$B = \frac{\sum_{i=1}^{n} x_i\left(c_i - \bar{c}\right)}{\sum_{i=1}^{n}\left(c_i - \bar{c}\right)^2} \tag{C.3}$$

where

$\bar{c}$ is the average of the c values, i.e. the average of the reference material values;

c_i is the individual reference material value.

Secondly the function $x_i = a + B\left(c_i - \overline{c}\right)$ is converted to $x_i = A + B\, c_i$ through the calculation of A according to Equation (C.4):

$$A = a - B\,\overline{c} \tag{C.4}$$

C.3 Calculation of the residuals of the average concentrations

The residuals of the average concentration at each concentration level to the regression line are calculated as follows.

Calculate at each concentration level the average of the AMS readings at one and the same reference material level c according to Equation (C.5):

$$\overline{x}_c = \frac{1}{m_c}\sum_{i=1}^{m_c} x_{c,\,i} \tag{C.5}$$

where

$\overline{x}_c$ is the average AMS reading at reference material level c;

$x_{c,\,i}$ is the individual AMS reading at reference material level c;

m_c is the number of repetitions at one and the same reference material level c.

Calculate the residual d_c of each average according to Equation (C.6):

$$d_c = \overline{x}_c - \left(A + Bc\right) \tag{C.6}$$

Convert d_c into a relative residual $d_{c,\mathrm{rel}}$ according to Equation (C.7) by dividing d_c by the upper limit of the certification range x_u:

$$d_{c,\,\mathrm{rel}} = \frac{d_c}{x_\mathrm{u}} \times 100\ \% \tag{C.7}$$

Annex D
(normative)

Determination of the total uncertainty

D.1 Determination of uncertainty contributions

The individual standard uncertainties, the combined standard uncertainty and the expanded uncertainty shall be determined according to the requirements of EN ISO 14956.

The individual standard uncertainties caused by the relevant performance characteristics shall be calculated by use of the maximum deviations or maximum standard deviations determined for the two AMS tested. This provides a worst-case estimate of the total uncertainty in the performance test.

D.2 Elements required for the uncertainty determinations

Table D.1 shows the uncertainty contributions which are to be combined when determining the combined standard uncertainty u_c. Specific contributions depend on the type of AMS, although some are common to all types of AMS.

Table D.1 — Uncertainty contributions

Number i	**Performance characteristic**	**Uncertainty** u_i
1	Lack of fit	u_{lof}
2	Zero drift from field test	$u_{d,z}$
3	Span drift from field test	$u_{d,s}$
4	Influence of ambient temperature at span	u_t
5	Influence of sample gas pressure	u_p
6	Influence of sample gas flow	u_f
7	Influence of supply voltage	u_v
8	Cross-sensitivity (interference)	u_i
9	Repeatability standard deviation at span [a]	$u_r = s_r$
10	Standard deviation from paired measurements under field conditions [a]	$u_D = s_D$
11	Uncertainty of the reference material provided by the manufacturer [b]	u_{rm}
12	Excursion of measurement beam [b]	u_{mb}
13	Converter efficiency for AMS measuring NO_x [b]	u_{ce}
14	Variation of response factors (TOC) [b]	u_{rf}

a Either the repeatability standard deviation at span or the standard deviation from paired measurements under field conditions is used, whichever is the larger.

b This uncertainty contribution is relevant for specific AMS only.

If the AMS depends on the regular use of reference materials for its continued accurate and precise operation – for example, during weekly span checks – then the uncertainty of the reference materials shall be included within the calculation of the total uncertainty.

The combined standard uncertainty u_c shall be determined by use of Equation (D.1) and summation of the uncertainty contributions u_i of the relevant performance characteristics specified in Table D.1:

$$u_c = \sqrt{\sum_{i=1}^{N} u_i^2} \tag{D.1}$$

The total expanded uncertainty U shall be determined using Equation (D.2):

$$U = 1{,}96\ u_c \tag{D.2}$$

In the above calculation, most the values of u_i for a parameter i are determined from test data, where the probability distribution of values is rectangular for most parameters and a normal distribution for a few parameters. Factor 1,96 may be used since the number of measurements to determine the uncertainty contribution and the associated number of degrees of freedom is sufficiently high or a rectangular distribution is assumed.

In the case of rectangular distributions the standard uncertainties for each performance characteristic are calculated according to EN ISO 14956 by Equation (D.3):

$$u_i = \sqrt{\frac{(x_{i,\text{max}} - x_{i,\text{adj}})^2 + (x_{i,\text{min}} - x_{i,\text{adj}}) \times (x_{i,\text{max}} - x_{i,\text{adj}}) + (x_{i,\text{min}} - x_{i,\text{adj}})^2}{3}} \tag{D.3}$$

where

$x_{i,\text{min}}$ is the minimum value of the average reading influenced by performance characteristic i during the performance test;

$x_{i,\text{max}}$ is the maximum value of the average reading influenced by performance characteristic i during the performance test;

$x_{i,\text{adj}}$ is the value of the average reading with the influence quantity at its nominal value during the performance test.

Equation (D.3) can be simplified in the following three cases:

— if the value $x_{i,\text{adj}}$ is at the centre of the interval bounded by the maximum value $x_{i,\text{max}}$ and the minimum value $x_{i,\text{min}}$ of all values x_i, then the standard uncertainty u_i is given by Equation (D.4):

$$u_i = \frac{(x_{i,\text{max}} - x_{i,\text{min}})}{\sqrt{12}} \tag{D.4}$$

— if the absolute values of the measured deviation above and below the central value are equal (see Equation (D.5)), then the standard uncertainty u_i is given by Equation (D.6):

$$\left| x_{i,\text{max}} - x_{i,\text{adj}} \right| = \left| x_{i,\text{min}} - x_{i,\text{adj}} \right| = \Delta x_i \tag{D.5}$$

$$u_i = \frac{\Delta x_i}{\sqrt{3}} \tag{D.6}$$

— if the value of $x_{i,\text{adj}}$ is the same as either $x_{i,\text{min}}$ or $x_{i,\text{max}}$, then the standard uncertainty u_i is given by Equation (D.7):

$$u_i = \frac{(x_{i,\text{max}} - x_{i,\text{min}})}{\sqrt{3}} \tag{D.7}$$

D.3 Example of an uncertainty calculation for a SO_2 AMS

An AMS installed at a refinery combustion process measures SO_2 and has a certified range of 0 mg/m^3 to 200 mg/m^3. The process falls under the Large Combustion Plant Directive and has an ELV of 100 mg/m^3 expressed as a daily average. The zero reading is checked automatically each day whilst span checks are carried out once per week. Table D.2 shows the performance characteristics of the AMS.

Table D.2 — Performance characteristics of an AMS for the uncertainty calculation

Performance characteristic	Test data
Certified range of the AMS	0 mg/m^3 to 200 mg/m^3
Test gas concentration	150 mg/m^3 ± 2,0 %
Lack of fit	±0,7 % of range
Zero drift in the maintenance interval	±1,0 % of range
Span drift in the maintenance interval	±2,7 % of range
Influence of ambient temperature at span	±1,0 % of range
Influence of sample gas pressure	±0,4 % of range
Influence of sample gas flow at 10 l/h	±2,0 % of range
Influence of supply voltage	±0,12 % of range
Cross-sensitivity	±2,1% of range
Repeatability standard deviation at span	0,8 % of range
Standard deviation from paired measurements under field conditions	0,65 % of range

The deviations of the measured signals obtained in the performance testing of the individual performance characteristics are converted to standard uncertainties on the basis of rectangular distributions. The test results in Table D.2 represent the worst-case values from the two AMS tested and correspond to the maximum variation of the corresponding influence quantities.

The AMS depends on the test gas for continued quality assurance and control. Therefore, the uncertainty of the test gas is included in the uncertainty calculations.

The repeatability standard deviation at span is used in the calculations, since it is greater than the standard deviation from paired measurements under field conditions.

Table D.3 shows the uncertainty calculations for each applicable performance characteristic.

Table D.3 — Uncertainty calculations

Performance characteristic	**Standard uncertainty**	**Value of the standard uncertainty** mg/m^3	**Square of the standard uncertainty** (mg/m^3)^2
Lack of fit	u_{lof}	$\frac{(0{,}7/100)\times 200}{\sqrt{3}} = 0{,}81$	0,656 1
Zero drift	$u_{\mathrm{d,z}}$	$\frac{1{,}0\ \%\times 200}{\sqrt{3}} = 1{,}15$	1,322 5
Span drift	$u_{\mathrm{d,s}}$	$\frac{2{,}7\ \%\times 200}{\sqrt{3}} = 3{,}12$	9,734 4
Influence of ambient temperature at span	u_{t}	$\frac{1\ \%\times 200}{\sqrt{3}} = 1{,}15$	1,322 5
Influence of sample gas pressure	u_{p}	$\frac{0{,}4\ \%\times 200}{\sqrt{3}} = 0{,}46$	0,211 6
Influence of sample gas flow	u_{f}	$\frac{2\ \%\times 200}{\sqrt{3}} = 2{,}31$	5,336 1
Influence of supply voltage	u_{v}	$\frac{0{,}12\ \%\times 200}{\sqrt{3}} = 0{,}14$	0,019 6
Cross-sensitivity	u_{i}	$\frac{2{,}1\ \%\times 200}{\sqrt{3}} = 2{,}43$	5,904 9
Repeatability standard deviation at span	u_{r}	$0{,}8\ \%\times 200 = 1{,}60$	2,560 0
Uncertainty of test gas	u_{tg}	$\frac{2\ \%\times 150}{\sqrt{3}} = 1{,}73$	2,992 9
Sum	–	–	30,060 6

The combined uncertainty is calculated using Equation (D.8):

$$\begin{aligned} u_{\mathrm{c}} &= \sqrt{u_{\mathrm{lof}}^2 + u_{\mathrm{d,z}}^2 + u_{\mathrm{d,s}}^2 + u_{\mathrm{t}}^2 + u_{\mathrm{p}}^2 + u_{\mathrm{f}}^2 + u_{\mathrm{v}}^2 + u_{\mathrm{i}}^2 + u_{\mathrm{r}}^2 + u_{\mathrm{tg}}^2} \\ &= \sqrt{30{,}0606\ (\mathrm{mg/m^3})^2} \\ &= 5{,}5\ \mathrm{mg/m^3} \end{aligned} \tag{D.8}$$

The expanded uncertainty at a level of confidence of 95 % is then calculated using Equation (D.9):

$$\begin{aligned} U_{0,95} &= 1{,}96 \times u_{\mathrm{c}} \\ &= 1{,}96 \times 5{,}5 \text{ mg/m}^3 \\ &= 10{,}8 \text{ mg/m}^3 \end{aligned} \tag{D.9}$$

The Large Combustion Plant Directive specifies an allowable expanded uncertainty of 20 % of ELV at 95 % confidence. As the ELV is 100 mg/m^3, the allowable expanded uncertainty is 20 mg/m^3. The maximum allowable expanded uncertainty in the suitability test is 75 % of 20 mg/m^3. Therefore, the AMS is compliant with these requirements in the performance test.

D.4 Determination of uncertainty contributions by use of sensitivity coefficients

The contribution of a variation of an influence quantity to the total uncertainty of the measured values in a future application of an AMS can be calculated from the range of values of the influence quantity in the considered application of the AMS and the sensitivity coefficient of this influence quantity determined in the performance test by use of Equation (D.10):

$$u_i = b_i \, u(X_i) \tag{D.10}$$

where

- u_i is the uncertainty contribution to the total uncertainty of the measured values caused by a variation of an influence quantity X_i;
- b_i is the sensitivity coefficient of the influence quantity X_i;
- $u(X_i)$ is the variation of the influence quantity X_i expressed as a standard uncertainty.

The variation of the influence quantity X_i can be converted to a standard uncertainty by use of the Equations (D.4) to (D.7).

D.5 Example of a calculation of an uncertainty contribution by use of a sensitivity coefficient

It is assumed that the sensitivity coefficient b_{t} = 0,1 (mg/m^3)/K of the ambient temperature influence has been determined in a performance test and that the AMS is installed in an environment with temperature variations between 5 °C and 40 °C about the nominal temperature of 20 °C. The temperature variation is converted to a standard uncertainty on the basis of Equation (D.3) by use of Equation (D.11):

$$\begin{aligned} u(T) &= \sqrt{\frac{(T_{\mathrm{max}} - T_{\mathrm{adj}})^2 + (T_{\mathrm{min}} - T_{\mathrm{adj}}) \times (T_{\mathrm{max}} - T_{\mathrm{adj}}) + (T_{\mathrm{min}} - T_{\mathrm{adj}})^2}{3}} \\ &= \sqrt{\frac{(313\,\mathrm{K} - 293\,\mathrm{K})^2 + (278\,\mathrm{K} - 293\,\mathrm{K}) \times (313\,\mathrm{K} - 293\,\mathrm{K}) + (278\,\mathrm{K} - 293\,\mathrm{K})^2}{3}} \\ &= \sqrt{\frac{(20\,\mathrm{K})^2 + (15\,\mathrm{K}) \times (20\,\mathrm{K}) + (15\,\mathrm{K})^2}{3}} \\ &= \sqrt{\frac{325\,\mathrm{K}^2}{3}} \\ &= 10{,}4\,\mathrm{K} \end{aligned} \tag{D.11}$$

The uncertainty contribution to the total uncertainty of the measured values caused by assumed variations in the ambient temperature is calculated by Equation (D.12):

$$\begin{aligned} u_{\mathrm{t}} &= b_{\mathrm{t}}\ u(T) \\ &= 0{,}1\ \frac{\mathrm{mg}}{\mathrm{m}^3}/\mathrm{K} \times 10{,}4\ \mathrm{K} \\ &= 1{,}0\ \frac{\mathrm{mg}}{\mathrm{m}^3} \end{aligned} \tag{D.12}$$

Annex E
(informative)

Elements of performance testing report

1. General

1.1 Certification proposal

1.2 Unambiguous AMS designation

1.3 Measured component(s)

1.4 Device manufacturer together with full address

1.5 Field of application

1.6 Measuring range for suitability test

1.7 Restrictions

Restrictions shall be formulated if testing shows that the AMS does not cover the full scope of possible application fields.

1.8 Notes

In the event of supplementary or extended testing, reference shall be made to all preceding test reports. Attention shall de drawn to main equipment peculiarities.

1.9 Test laboratory

1.10 Test report number and date of compilation

2 Task definition

2.1 Nature of test

First test or supplementary testing.

2.2 Objective

Specification of which performance criteria were tested;

Bibliography;

Scope of any supplementary tests.

3 Description of the AMS tested

3.1 Measuring principle

Description of metrological and scientific relationships.

3.2 AMS scope and set-up

Description of all parts of the AMS covered in the scope of testing, if possible including a copy of an illustration or flow diagram showing the AMS. Statement of technical specifications, if appropriate in tabular form.

4 Test program

Details shall be provided on the test program, in relation to the AMS under test.

In the case of supplementary or extended testing, the additional scope of testing shall be detailed and substantiated.

Particularities of the test shall be documented.

4.1 Laboratory test/laboratory inspection

Statement of all test steps involved

4.2 Field test

Details on:

- all test steps involved;

- plant type on which the field test examinations were carried out;

- AMS measuring range to be covered in the test;

- installation conditions and operating conditions for the AMS under test.

5 Standard reference method

5.1 Method of measurement

It is necessary to specify the standard reference method employed. Variations from any method acknowledged as Standard Reference Method of measurement as described in European, international or national standards shall be documented. Only validated methods shall ever be used and, as such, a statement on validation shall be made. If, in substantiated cases of exception, continuous AMS are used, details shall be provided on the analyser type, manufacturer, measuring range selected and the suitability test applicable to this equipment type. The standard deviation of the standard reference method shall be stated.

5.2 Test stand set-up

Description of the sampling probe, any dust filters used for particle separation in the measurement of gaseous substances, details on the sample gas line (length, material, size) and on sample gas conditioning.

If continuous AMS are used, the AMS characteristic shall first be examined in accordance with the specifications given in EN 14181. The certified test gases used for this purpose shall be described in respect of their specifications.

6 Test results

Comparison of the performance criteria placed on continuous emission AMS in the performance test with the results attained.

The information below shall be stated for each individual test point in the following order of sequence:

Consecutive number and short title of performance criteria as heading.

6.1 Citation of performance criterion

6.2 Equipment

6.3 Method

6.4 Evaluation

6.5 Assessment

6.6 Detailed presentation of test results allowing for the respective section of the documentation

Annex A Values measured and computed

Annex B Operating instructions

The operating instructions should also be enclosed with the report in electronic form (e.g. as a PDF file).

Bibliography

[1] Directive 2001/80/EC of the European Parliament and of the Council of 23 October 2001 on the limitation of emissions of certain pollutants into the air from large combustion plants, OJ L 309, p. 1

[2] Directive 2000/76/EC of the European Parliament and of the Council of 4 December 2000 on the incineration of waste, OJ L 332, p. 91

[3] Council Directive 1999/13/EC of 29 March 1999 on the limitation of emissions of volatile organic compounds due to the use of organic solvents in certain activities and installations, OJ L 85, p. 1

[4] Council Directive 96/61/EC of 24 September 1996 on Integrated Pollution Prevention and Control, OJ L 257, p. 26

[5] Directive 2003/87/EC of the European Parliament and of the Council of 13 October 2003 covering processes emitting greenhouse gases, OJ L 275, p. 32

[6] EN ISO 9169, *Air quality — Definition and determination of performance characteristics of an automatic measuring system (ISO 9169:2006)*

[7] ISO 5725-2, *Accuracy (trueness and precision) of measurement methods and results — Part 2: Basic method for the determination of repeatability and reproducibility of a standard measurement method*

[8] CEN/TS 15675, *Air quality — Measurements of stationary source emissions — Application of EN ISO/IEC 17025:2005 to periodic measurements*

[9] ENV 13005:1999, *Guide to the expression of uncertainty in measurement (GUM)*

[10] International Vocabulary of basic and general terms in metrology (VIM), BIPM, IEC, IFCC, ISO, IUPAC, IUPAP, OIML, 1993

blank

BSI — British Standards Institution

BSI is the independent national body responsible for preparing British Standards. It presents the UK view on standards in Europe and at the international level. It is incorporated by Royal Charter.

Revisions

British Standards are updated by amendment or revision. Users of British Standards should make sure that they possess the latest amendments or editions.

It is the constant aim of BSI to improve the quality of our products and services. We would be grateful if anyone finding an inaccuracy or ambiguity while using this British Standard would inform the Secretary of the technical committee responsible, the identity of which can be found on the inside front cover. Tel: +44 (0)20 8996 9000. Fax: +44 (0)20 8996 7400.

BSI offers members an individual updating service called PLUS which ensures that subscribers automatically receive the latest editions of standards.

Buying standards

Orders for all BSI, international and foreign standards publications should be addressed to Customer Services. Tel: +44 (0)20 8996 9001. Fax: +44 (0)20 8996 7001. Email: orders@bsi-global.com. Standards are also available from the BSI website at http://www.bsi-global.com.

In response to orders for international standards, it is BSI policy to supply the BSI implementation of those that have been published as British Standards, unless otherwise requested.

Information on standards

BSI provides a wide range of information on national, European and international standards through its Library and its Technical Help to Exporters Service. Various BSI electronic information services are also available which give details on all its products and services. Contact the Information Centre. Tel: +44 (0)20 8996 7111. Fax: +44 (0)20 8996 7048. Email: info@bsi-global.com.

Subscribing members of BSI are kept up to date with standards developments and receive substantial discounts on the purchase price of standards. For details of these and other benefits contact Membership Administration. Tel: +44 (0)20 8996 7002. Fax: +44 (0)20 8996 7001. Email: membership@bsi-global.com.

Information regarding online access to British Standards via British Standards Online can be found at http://www.bsi-global.com/bsonline.

Further information about BSI is available on the BSI website at http://www.bsi-global.com.

Copyright

BSI
389 Chiswick High Road
London
W4 4AL